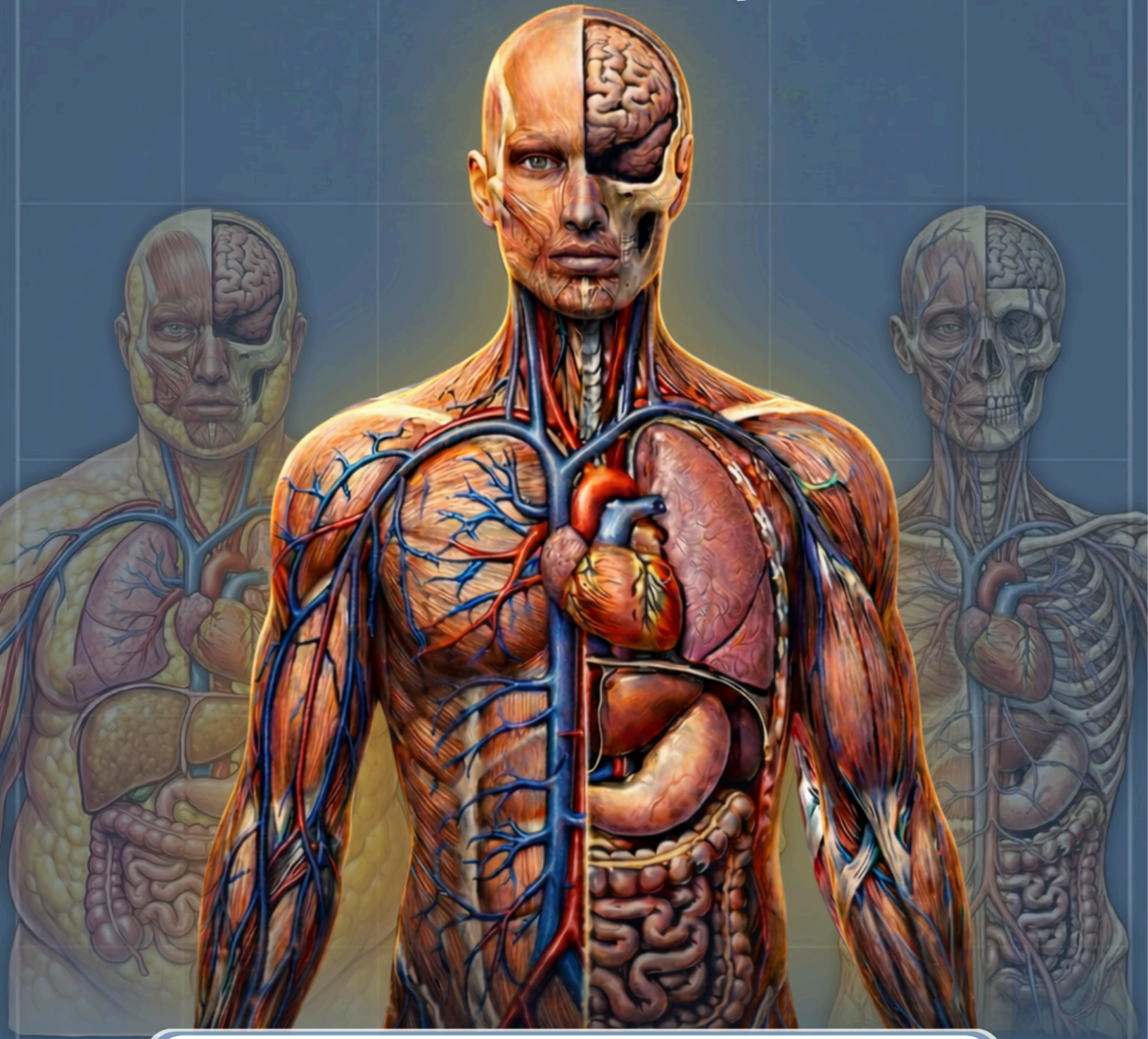


PEPTIDES

Optimize Your Body & Mind



World Peptide Association

Important Notice and Disclaimer: For Research Use Only

It is critical for all readers, users, and customers to understand that all products referenced, described, or discussed within this book, and potentially on the affiliated website, worldpeptideassociation.com, are strictly and unequivocally designated for Research and Development (R&D) purposes only.

These compounds are raw chemical materials intended solely for *in vitro* (test tube) and *ex vivo* (outside a living organism) scientific investigations, laboratory experimentation, and professional research applications.

Under no circumstances whatsoever are these products intended or approved for Human consumption or administration of any kind. This includes, but is not limited to, oral ingestion, injection, topical application, or any other method of introduction into the human body. The misuse of these research-grade materials constitutes a violation of this disclaimer and is entirely at the user's risk.

Disclaimer on Health and Medical Efficacy:

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Why WPA and this book?

The Definitive Book for the Research Peptide Community

The World Peptide Association (WPA) stands alone as the paramount authority and indispensable resource for the global research peptide community. Our mission is to foster knowledge, ensure safety, and provide unparalleled data transparency to researchers operating in the *for research purposes* domain. This book is a direct extension of that commitment, providing the most current, data-driven insights available anywhere. The Uniqueness of WPA Data: The WPA Calculator

What truly sets the WPA apart is our exclusive and robust data source for understanding the real-world methodologies employed by research professionals: The WPA Calculator. This tool is not just a mechanism for planning; it is the world's only continuous, dynamic, and rapidly growing data collection platform focused on the self-reported practices of the research peptide community.

We gather critical, high-fidelity information anonymously, allowing researchers to contribute their real-world usage protocols without compromising their privacy. This self-reported data includes:

1. What Compound Is It?

- **Name of Compound:** Clearly state the specific peptide or chemical you are working with.

2. Mixing Instructions (Your 'Recipe')

- **Vial Size (mg):** How much of the pure powder is in the sealed bottle? (The starting amount).
- **Water Added (mL):** How much water (specifically, "Bac Water") did you add to the

powder? (This is key for figuring out the strength).

3. Dosing & Schedule

- **Target Dose (mcg or mg):** How much of the mixture do you plan to inject each time? (Be specific to avoid errors).
- **How Often:** How many times a week do you plan to inject it?

This data—compound identity, starting mass, reconstitution volume, target dose, and dosing schedule—is the basis for calculating concentration, injection volume, and vial longevity.

No other organization on the planet has access to this depth or breadth of current research practice data, nor are they collecting it with the speed and commitment of the WPA. Commitment to Transparency and Value

As of the preparation of this definitive guide, the WPA Calculator is rapidly approaching 20,000 individual calculations—a staggering repository of self-reported research data. The WPA does not hoard this information; instead, we freely analyze and share this aggregated data to bring essential transparency, credibility, and value to the community we serve.

This book leverages that monumental dataset, transforming raw practice statistics into actionable knowledge, establishing best practices, and ensuring that researchers have access to the most informed, community-validated information available. It is designed to be the authoritative manual, powered by the collective experience of the world's peptide researchers.

How to Use the WPA Peptide Calculator (Step-by-Step)

These steps mirror the workflow on worldpeptideassociation.com so researchers can quickly translate a protocol into a syringe draw volume.

The Smart Assistant

- 1 Select Your Compound**
Search for your peptide by name or browse popular options.
Tirzepatide
Retatrutide
Semaglutide
- 2 Enter Vial Size (mg)**
This is printed on your vial's label — the total amount of peptide powder.
10 mg
20 mg
5 mg
- 3 Enter BAC Water (mL)**
How much bacteriostatic water you'll add to reconstitute the vial.
2 mL
3 mL
1 mL
- 4 Enter Desired Dose**
Type the amount you want per dose and select MCG or MG.
Don't forget to select MCG or MG for your dose unit.

For research purposes only.

Click the graphic above to launch the WPA Peptide Calculator

1. Choose your peptide.
2. Decide your research dose for a single administration (mcg and mg are shown throughout this guide).
3. Enter your vial's total content (e.g., 10 mg).
4. Pick BAC water volume for reconstitution. You can choose any volume; the calculator also shows community analytics for commonly used volumes.
5. Click Calculate Dosage. The syringe graphic highlights the draw line corresponding to your dose with the chosen dilution.
6. Record your dose → draw volume for the protocol schedule.

Tip: If the draw volume seems too small or too large, adjust the BAC water volume and recalculate until the draw is at a level you are comfortable with.

WPA Data Advantage: Truth Over "Bro Science"

The Problem: Most research protocols come from a game of "telephone." When researchers are going to start researching with a new compound they might turn to a forum, reddit or youtube or see what a friend is taking, which is sometimes based on some form of research in animal models that may or may not translate to humans. Someone posts on a forum, an influencer repeats it, and suddenly it becomes a "rule." But is it actually correct? Or is it just Bro Science?

The Solution: We don't guess. We count. **Driving Data-Informed Research Practices.** Every instance of the WPA Calculator's use is an opportunity to anonymously enhance a growing, comprehensive dataset on dosing practices. This collection is meticulous, automatically recording essential, anonymous parameters that allow us to observe real-world trends.

The key data points we anonymously collect are:

1. **What compound?** Tracking the specific substances being prepared provides insight into the most commonly researched compounds and helps researchers understand which compounds are the best for their researcher.
2. **What vial size?** If someone wants to start researching a new compound they are easily able to see what the most common vial size is based on thousands of fellow researchers.
3. **How much BAC water?** The volume of Bacteriostatic Water used as a diluent (fancy word for mixing) is essential and WPA analytics show you exactly how much BAC water for every compound.
4. **What dose?** The target dose is the most critical variable, offering a direct view into the range of dosages researchers are aiming for.
5. **How often?** The frequency of administration provides context for experimental design and total cumulative substance use over a defined period.

By aggregating and analyzing this secure, anonymous usage data, we can identify best practices, flag potential areas of confusion, and continuously refine the WPA Calculator's algorithms and user experience. This data-driven approach ensures that the tool remains the most accurate, reliable, and relevant resource for standardized and safe dosing preparation in the research community.

With nearly **20,000 unique calculations** (and growing daily), we have built the world's largest dataset on real-world peptide usage. This allows us to ignore the "noise" and show you the **Ground Truth**.

The Macro View (Industry Trends)

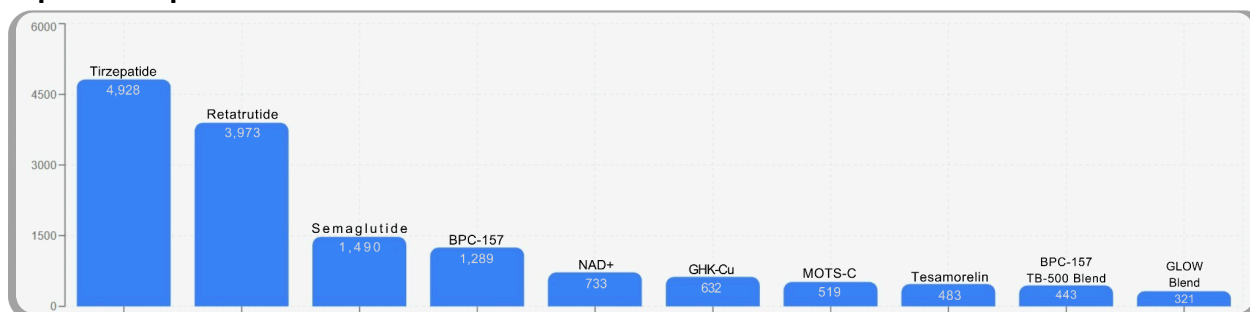
First, let's look at the big picture. What is the rest of the world actually researching?

1. The Popularity Contest (All-Time Leaders)

You don't need to ask "What's popular?" The data tells us instantly.

- **The Insight:** While everyone talks about the "new" stuff, the data shows that the classics (like Tirzepatide and BPC-157) still dominate the landscape.

Top 10 Compounds Bar Chart



The Leaderboard: Based on nearly 20,000 data points, this chart shows exactly which compounds are being researched the most right now - [click here](#) to see up to date data.

2. The "Trending" List (What's Hot Now)

Usage shifts fast. We track momentum over the last 30 days to see what is exploding in popularity and what is fading away.

- **The Insight:** Use this to spot the "Next Big Thing" before it hits the forums.

Trending/Gaining Popularity

- **What to Capture:** The list showing "Tirzepatide +44%", "Retatrutide +57%", etc.

"The Movers & Shakers: Green means growing. Red means slowing. See [click here](#) which compounds are capturing the community's attention this month."

3. The Mixing Standard (BAC Water)

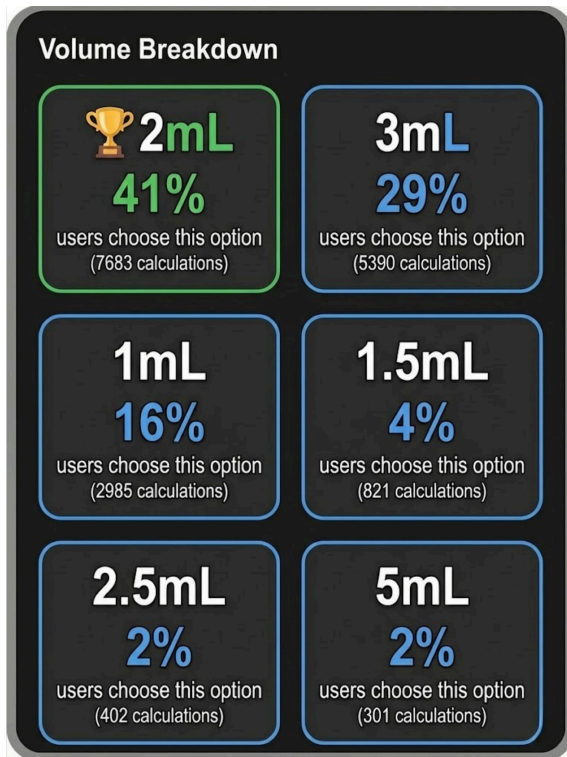
"How much water should I add?" This is the #1 question for beginners.

Which Compounds Are Gaining Popularity?

Compounds ranked by how many calculations researchers have over the last 30 days. The **green** news growing **interest** compared 30 day. (▲▼) indicate wv positions a compound has moved up or down. 2 compounds also calculations from the past 24 hours.

- 1 Tirzepatide ~44%
- 2 Retatrutide ~57%
- 3 Semaglutide ~157% ▲4
- 4 BPC-157 ~25% ▼1
- 5 NAD+ ~70%
- 6 GHK-Cu ~82%
- 7 MOTS-c ~28% ▲1
- 8 Tesamorelin ~68% ▲2
- 9 GLOW Blend ~141% ▲5
- 10 Ipamorelin ~47% ▲2

- **The Truth:** There is no "wrong" answer, but there is a "common" answer. Our data shows that while **41%** of users prefer **2mL**, a solid **29%** still prefer **3mL**.



BAC Water Usage ChartLink:

The Consensus: Don't stress about the water. As you can see [click here](#), preferences vary. Use this chart to see what the majority of researchers are doing.

The Micro View (Compound Specifics)

Now, let's zoom in. Let's look at **Tirzepatide** as an example of how granular we can get.

4. The "Real" Dose

If you ask Reddit, they might say "Everyone starts

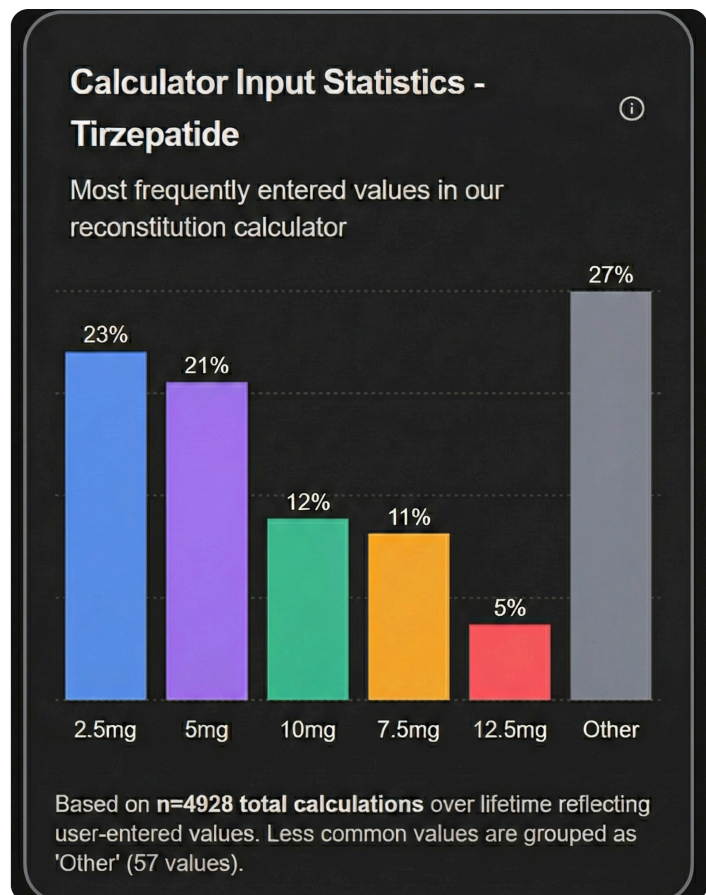
at 2.5mg." But is that true?

- **The Data:** It's nuanced. Our data reveals a near-even split: **23%** of researchers stick to the starter dose (2.5mg), while **21%** have moved up to 5mg.
- **Why it matters:** Seeing the distribution helps you realize there is no "One Size Fits All."

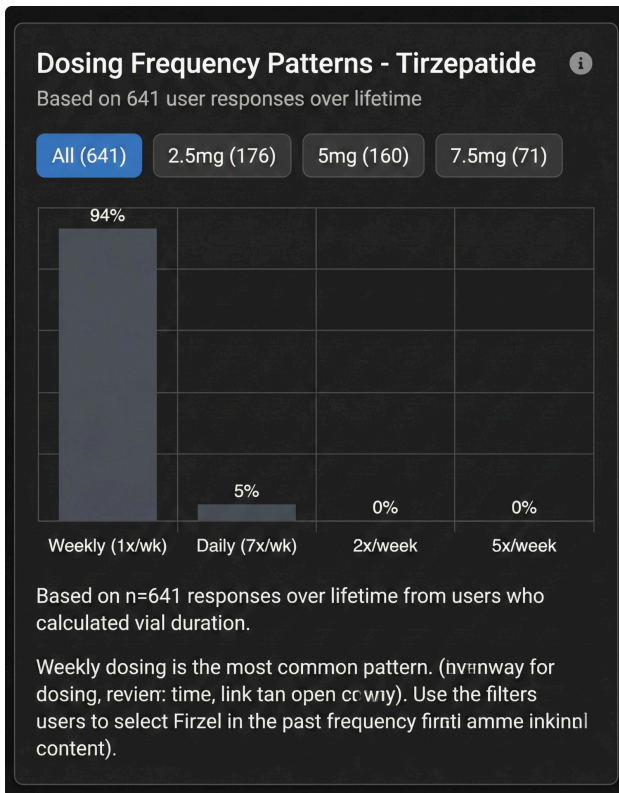
Tirzepatide Dosing Amounts

- **What to Capture:** The bar chart showing the breakdown of specific doses (2.5mg, 5mg, 7.5mg, etc.).

Benchmarks, Not Guesses: See exactly how many users are at your dose level versus moving up [click here](#). This helps



you benchmark your research against the 'Norm'.



5. Frequency Patterns

How often should you research?

- **The Standard:** For Tirzepatide, the data confirms the standard protocol: **94%** of users administer it **Once Weekly**.
- **The Outliers:** Interestingly, **5%** of users are experimenting with "Micro-dosing" (7 days a week). The data captures these emerging protocols that you might otherwise miss.

Dosing Frequency Pie Chart

- **Caption:** "Protocol Check: Are you an outlier? [This chart](#) shows how frequently the majority of researchers administer this specific compound."

WPA Verified Vendors

The "Wild West" Problem In the research peptide market, not everyone plays by the rules. The most critical safety check is the **COA (Certificate of Analysis)**. A proper COA tests for two things:

1. **Purity:** Is the product actually 99%+ pure, or is it full of who knows what?
 2. **Assay (Quantity):** If the vial says "10mg," is there actually 10mg inside, or did they only put 5mg to save money?
-

Step 1: The Filter (Finding Safe Products)

You don't have to guess. Our tools allow you to filter the entire market to find exactly what you need. You can sort by **Recency** (to avoid the "Dusty Scenario") and decide what "Value" means to you:

- **Price Per Vial:** Great for researchers on a budget who just need one bottle.
- **Price Per MG:** The smart way to buy in bulk. Often, a more expensive vial is actually a better deal because it has double the product.

Tirzepatide – compare research peptide offers

Each row represents a vendor and each dot represents a vial size. Prices include WPA discounts where available.

☒ \$ / mg ☐ Total vial price ☐ COA purity % ☐ Assay mg/vial

Vial size: COA: COA date: Stock:

[Click Here](#) to Use This Tool: Use the live filters to instantly hide any product that hasn't been tested recently. You can also toggle between "Price Per Vial" and "Price Per MG" to find the best setup for your research.

Step 2: The Vendor Quality Check

Transparency is everything. We list every vendor we track, even the ones who don't have their paperwork in order. This allows you to see the difference side-by-side.

The COA "Freshness" (Why Dates Matter) Checking for a COA isn't enough—you have to check the **Date**. At WPA, we recommend looking for tests performed within the last **6 months**.

Why? Because an old COA (e.g., from 3 years ago) usually implies one of three bad things:

1. **The "Lazy" Scenario:** They are making new batches but aren't bothering to test them (so you have no idea what you are buying).
2. **The "Ghost" Scenario:** They are testing new batches, failing, and hiding the results.
3. **The "Dusty" Scenario:** They are still selling the *same* batch from 3 years ago, which means the product has likely degraded from sitting on a shelf for too long.

None of these are good options.

Vendor ↑↓	Shop	Popularity (90d)	Price		COA DATA		
			\$/mg ↑↓	\$/vial ↑↓	Date ↑↓	Purity ↑↓	mg/vial ↑↓
SOMA 72mg	Save \$97.20 (16% Off) Code: WPA Shop Vendor	Low volume 0.0	\$7.65/mg	\$550.80	Missing	Missing	Missing
Ascension Peptides 10mg	Save \$30.00 (20% Off) Code: WPA Shop Vendor	Low volume ↓ 1.0	\$12.00/mg	\$119.99	7/24	99.7%	Missing
Apollo Peptide Sciences 100mg	Save \$70.00 (10% Off) Code: WPA Shop Vendor	Low volume ↓ 1.1	\$6.30/mg	\$629.99	2/25	99.5%	102.5 mg

[Click Here](#) to See the Live Vendor List: *Don't fly blind, see which vendors currently have COAs and which ones are missing critical testing data.*

Step 3: Visualizing Value (The Dot Chart)

Sometimes a list of numbers is hard to read. That is why we built the **Dot Chart**. The dot chart is visible when you are looking for any product on WPA sites. Sometimes a list of 40 different options of the same product can be overwhelming, that is why we built the dot chart.

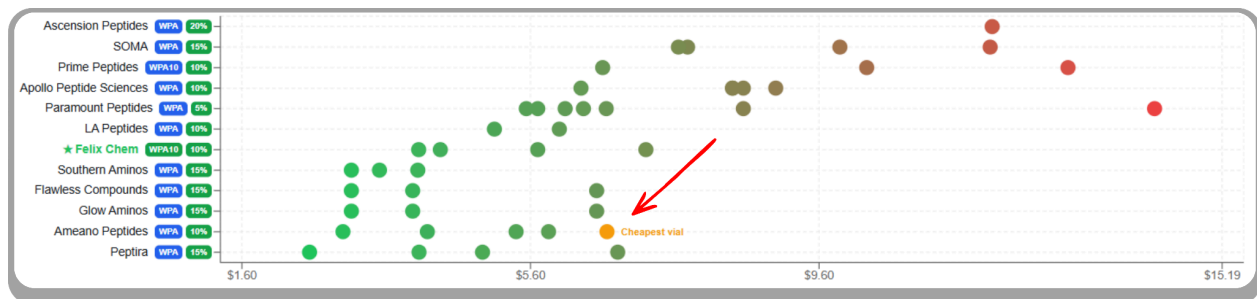
- **The Dot Chart:** Each dot is a different offer from a different company. The chart allows sorting by both **Price/Vial** and **Price/mg** along with several other ways (see above) to

satisfy two different buyer needs.

- **Research group 1:** Wants the cheapest priced vial, we highlight the yellow dot to make picking your next research product easy.
- **Price/mg (Horizontal Axis):** This helps the second group of researchers, who need larger quantities for longer studies, find the most cost-effective per-milligram price (often found in larger vials/quantities). The horizontal axis is color-coded: bright green dots on the left indicate the lowest Price/mg, while the transition to red on the far right indicates the highest Price/mg.

The Red Arrow: In the example below, the arrow points to the absolute lowest price on the market for this specific compound, representing the cheapest Price/Vial.

Note: The vertical axis also shows different vendors. Vendors may offer multiple vial sizes for the same product, and the chart helps identify any inconsistencies in their pricing.



[Click Here](#) to Find the Cheapest Deal: Stop overpaying! The ones on the left are the best deals. Click this chart to see who is charging fair prices right now.

How This Book Works

This book is designed for efficiency. Every compound entry follows a precise, identical structure so you can find critical research data in seconds.

1. The Anatomy of an Entry

Each peptide profile is broken down into five distinct sections:

- The Science ("*What it is / research focus*") A technical summary of the peptide's mechanism of action, listing primary targets and physiological effects based on clinical or preclinical literature.
- The Translation ("*In simple terms*") A plain-English summary aimed at 5th-grade readability. If the science gets too dense, look here for the "bottom line."
- The Real-World Context ("*Anecdotal*") Observations from independent researchers. This section highlights subjective effects (e.g., "improved sleep") or side effects that may not yet be documented in clinical trials.
- The Research Protocols ("*Beginner / Intermediate / Advanced*") Tiered dosing schedules designed to match research goals.
 - *Includes:* Specific dosage (mg and mcg), frequency, and cycle duration.
 - *Note:* Protocols move from conservative (safety-first) to aggressive (efficacy-first).
- The Support Stack ("*Supplements & diet*") Nutritional and supplemental adjuncts that may synergize with the peptide or mitigate specific side effects.
- The Safety Signal ("*Cautions*") A summary of known side effects, contraindications, or adverse events.
 - *Source:* Derived from both clinical literature and common anecdotal reports.
 - *Purpose:* To identify "red flags" and expected physiological responses during research.

2. Universal Research Standards

Unless explicitly noted otherwise in a specific entry, the following standards apply to every

compound in this book:

- Reconstitution: Assume Bacteriostatic Water (BAC water) is the standard solvent for all lyophilized powders.
- Administration: All protocols imply Subcutaneous (SubQ) administration. Intramuscular (IM) is rarely required unless specified.
- Storage:
 - *Powder (Unreconstituted)*: Store in a cool, dark place (freezer optional for long-term).
 - *Liquid (Reconstituted)*: Must be kept refrigerated (36-46°F). Avoid shaking or repeated warming.
- Evidence Hierarchy: This book prioritizes data in the following order:
Peer-Reviewed Human Trials → Animal/In-Vitro Studies → Independent Researcher Logs.

Pick A Goal Table of Context

Note: This isn't the whole menu! These are just the main categories. We cover many more compounds in the pages that follow.

Muscle & Performance Research

- [CJC-1295](#) / [Ipamorelin](#): Growth hormone secretion & recovery.
- [GHRP-2](#) / [GHRP-6](#) / [Hexarelin](#): Potent growth hormone secretion (appetite stimulation).
- [IGF-1 LR3](#) / [IGF-1 DES](#): Anabolic signaling & insulin-like growth factors.
- [PEG-MGF](#): Muscle tissue repair signaling.
- [Follistatin-315](#) / [Follistatin-344](#): Myostatin inhibition research.
- [5-Amino-1MQ](#): Cellular metabolism & muscle preservation.

Metabolic & Body Composition

- [Semaglutide](#) / [Tirzepatide](#) / [Retatrutide](#) / [Liraglutide](#): GLP-1 agonists for satiety & insulin sensitivity.
- [AOD-9604](#) / [hGH Frag 176-191](#): Lipolysis (fat breakdown) signaling.
- [Tesamorelin](#): Visceral adipose tissue reduction.
- [Adipotide](#): Adipose tissue vasculature targeting.
- [MOTS-c](#) / [SS-31](#): Mitochondrial energy & metabolic regulation.

Injury Repair & Tissue Regeneration

- [BPC-157](#): Tendon, ligament, & gut mucosa healing.
- [TB-500 \(Thymosin Beta-4\)](#): Cellular migration & tissue remodeling.
- [GHK-Cu](#): Skin repair, collagen synthesis, & anti-inflammatory signaling.
- [ARA-290](#): Neuropathic repair & anti-inflammatory pathways.
- [KPV](#): Skin & gut anti-inflammatory; antimicrobial defense.

Cognitive Function & Nootropics

- [Semax](#) / [N-Acetyl Semax](#): Focus, clarity, & BDNF expression.
- [Selank](#) / [N-Acetyl Selank](#): Anxiolytic (calming) & cognitive support.
- [Cortagen](#) / [Pinealon](#): Neuropeptide regulation & stress response.
- [P21](#) / [PE-22-28](#): Neurogenesis & neuroprotection.

Longevity & Cellular Health

- [Epithalon](#): Telomere length regulation & circadian rhythm.
- [Thymosin Alpha-1](#): Immune system modulation & T-cell support.
- [Glutathione](#) / [NAD+](#): Antioxidant status & cellular energy.
- [Humanin](#): Cytoprotection & mitochondrial health.

Sleep, Mood, & Vitality

- [DSIP](#): Delta-wave sleep induction.
- [PT-141 \(Bremelanotide\)](#): Libido & sexual arousal signaling.
- [Melanotan II](#): Melanogenesis (pigmentation) & libido.
- [Oxytocin](#): Social bonding & stress regulation.

Peptide Compound Directory

5-Amino-1MQ

What it is / research focus: 5-Amino-1MQ is technically a "small molecule," not a peptide, which means it can be taken as a capsule. It is a specific inhibitor of an enzyme called NNMT.

- **The Problem:** As we age, our fat cells start overproducing NNMT. This enzyme impairs fat burning and slows down muscle repair.
- **The Goal:** To block this enzyme, forcing the fat cells to shrink and the muscle cells to repair themselves.

How it works (The Parking Brake Story): Think of your metabolism as a car.

- **The Problem (NNMT):** As you get older, your body starts pulling the Parking Brake (NNMT) while you are trying to drive. You can press the gas (exercise) and cut the fuel (diet), but because the brake is on, the car won't move, and you just waste energy.
- **The Solution (5-Amino-1MQ):** This compound reaches down and releases the parking brake.
- **The Result:** Suddenly, the same amount of gas (exercise) makes the car go much faster. The engine stops struggling, and the fuel efficiency (fat burning) skyrockets.

The "Strange" Fact (The Muscle Miracle): Most "fat burners" eat away at muscle, but 5-Amino-1MQ does the opposite.

- **The Study:** In famous experiments, aged mice given this compound became stronger and had better muscle endurance even without exercising. It seems to wake up "Satellite Cells" (muscle stem cells) that usually go dormant in old age, allowing old muscles to heal like young muscles.

In simple terms: It is a "Metabolic Brake Release." It stops your fat cells from fighting against your weight loss and helps muscles stay young.

Anecdotal:

- **"The Oral Advantage":** Because this survives digestion, researchers love that they don't have to inject it. It is almost exclusively found as capsules.

- **"The Clean Energy":** Unlike caffeine or pre-workouts that make you jittery, users report a "background" energy. They simply don't get tired as fast during the day.
- **"The De-Bloat":** A very common report is a rapid drop in water weight and inflammation within the first 10 days.

Beginner Protocol:

- **Dose:** 50 mg
- **Frequency:** Daily (Oral Capsule).
- **Timing:** Take with breakfast (it can be stimulating, so avoid night dosing).
- **Cycle:** 4–8 weeks.

Intermediate Protocol:

- **Dose:** 100 mg
- **Frequency:** Daily.
- **Cycle:** 6–8 weeks.

Advanced Protocol:

- **Dose:** 150 mg
- **Frequency:** Daily.
- **Note:** Doses higher than 150 mg have not shown better results in research and can get very expensive.

Supplements & diet:

- **NAD+ Boosters:** Since 5-Amino-1MQ works by saving NAD+, adding an NAD+ precursor (like NMN or NR) is like adding fuel to the fire.
- **Methyl Donors:** Unlike other compounds that deplete your body's methylation (energy) pool, 5-Amino-1MQ actually saves it. However, taking a simple B-Complex vitamin is still recommended to keep the pathways running smoothly.

Cautions:

- **Insomnia:** If you take it after 2:00 PM, you might find yourself staring at the ceiling at midnight. It wakes up the brain's energy systems.
- **Headaches:** Mild headaches are reported in the first week as the body adjusts to the new metabolic rate.

Adipotide (FTPP)

What it is / research focus: Adipotide is a very different kind of research compound. Most diet aids work by making you feel full or speeding up your metabolism. Adipotide works by "starving" the fat cells directly.

How it works (The Garden Hose Story): Imagine your body fat is like a big garden. For the grass (fat) to grow, it needs water from a garden hose (blood vessels).

- **The Target:** Adipotide is like a smart clamp. It is designed to find *only* the garden hoses that feed your body fat.
- **The Action:** It clamps those hoses shut so no blood can get through.
- **The Result:** Without fresh blood, the fat cells cannot survive. They shrink and die away.

Why is this serious? Because the fat cells are actually dying inside the body, the body has to clean up the mess. This is a lot of work for the system, which is why researchers often report feeling "flu-like" symptoms or getting dehydrated.

In simple terms: It cuts off the blood supply to fat cells. The fat cells run out of food and die.

Anecdotal: This is considered a "heavy duty" researcher tool. It is known for working fast, but it is not comfortable. Many reports mention feeling tired or having a tummy ache while the fat is breaking down. **Drinking a lot of water is non-negotiable** to help the kidneys flush out the waste.

Beginner Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg)
- **Frequency:** 2 to 3 times per week (SubQ)
- **Cycle:** 4–6 weeks

Intermediate Protocol:

- **Dose:** 1.0 mg – 2.0 mg (1,000 mcg – 2,000 mcg)
- **Frequency:** 2 to 3 times per week
- **Cycle:** 6–8 weeks

Advanced Protocol:

- **Dose:** 2.0 mg – 3.0 mg (2,000 mcg – 3,000 mcg)
- **Frequency:** 2 to 3 times per week

- **Cycle:** 8 weeks max

Supplements & diet:

- **Hydration is #1:** You must drink extra water to help your kidneys.
- **Protein:** Keep protein high to protect your muscles.
- **Electrolytes:** Add these to your water to prevent headaches.

Cautions:

- **Kidney Stress:** Because the body is flushing out dead fat cells, the kidneys work overtime.
- **Side Effects:** Nausea and fatigue are common. If back pain occurs (kidney area), research usually stops immediately.

AICAR

What it is / research focus: AICAR is famous for being an "exercise mimic." It is a research chemical that tricks the body into thinking it is working out, even when it is resting.

How it works (The Battery Meter Story): Inside your cells, there is a sensor called AMPK.

Think of AMPK as your body's "Battery Meter."

- **Normal Exercise:** When you run or lift weights, your battery meter drops. Your body sees this and starts burning sugar and fat to charge it back up.
- **The Trick:** AICAR enters the cell and pulls a prank. It manually turns the "Low Battery" light on.
- **The Result:** Your body panics (in a good way) and starts burning fuel to recharge the battery—even if the subject is just sitting on the couch. It also starts building more "power plants" (mitochondria) in the muscles to prepare for this fake workout.

Why is this researched? Scientists are studying it to see if it can help people who cannot exercise (like the elderly or injured) stay fit. Athletes have studied it for endurance because it might help muscles run longer without getting tired.

In simple terms: It flips the "exercise switch" in your cells. It makes the body burn fat and build endurance without actually moving.

Anecdotal: This is not a "feel good" compound. Researchers often report that it feels like doing a hard workout. "Hot flashes," sweating, and feeling winded or tired (even while sitting still) are common reports because the metabolism is working so hard.

Beginner Protocol:

- **Dose:** 5 mg – 10 mg
- **Frequency:** Daily (SubQ)
- **Cycle:** 2–4 weeks

Intermediate Protocol:

- **Dose:** 10 mg – 20 mg
- **Frequency:** Daily
- **Cycle:** 4–6 weeks

Advanced Protocol:

- **Dose:** 20 mg – 30 mg
- **Frequency:** Daily
- **Cycle:** 6–8 weeks

Supplements & diet:

- **Carbs:** Since AICAR burns blood sugar, having some healthy carbs (like oatmeal or fruit) is important so blood sugar doesn't drop too low.
- **Hydration:** Drink electrolytes.
- **Creatine:** Often added to help the muscles handle the extra energy demand.

Cautions:

- **"The Drop":** Because it pushes sugar into muscles very fast, it can cause low blood sugar (hypoglycemia). Researchers usually keep a snack nearby.
- **Heart Rate:** Some reports note a slight increase in heart rate (similar to light exercise).

AOD-9604

What it is / research focus: A modified piece of the human growth hormone molecule, specifically designed to target fat loss. It was originally developed by scientists in Australia who wanted to find a drug that could cure obesity without the side effects of regular growth hormone.

Why is it called "hGH Frag 176-191"? Imagine the Human Growth Hormone (hGH) molecule is a long train with 191 cars.

- **The Whole Train (hGH):** Does many things—it makes muscles grow, it raises blood sugar, and it burns fat.
- **The Fragment (176-191):** Scientists discovered that the "fat burning" ability was only

located in the very last few cars of the train (numbers 176 to 191).

- **The Result:** They cut off just that tail end. This "Fragment" sends the signal to burn fat, but because the rest of the train is missing, it **cannot** make you grow or mess up your blood sugar.

In simple terms: Think of this as "Diet Growth Hormone." It is a tiny slice of the growth hormone molecule that focuses only on breaking down fat. It doesn't make muscles grow or raise insulin like the full version does.

Anecdotal: Researchers often report it is best used during a "cutting" phase (dieting). It is not a magic pill—it works best when the subject is already active. Reports of side effects are very rare, usually just mild stomach upset.

Beginner Protocol:

- **Dose:** 0.25 mg – 0.5 mg (250 mcg – 500 mcg)
- **Frequency:** Daily (usually in the morning on an empty stomach)
- **Cycle:** 8–12 weeks

Intermediate Protocol:

- **Dose:** 0.5 mg – 0.8 mg (500 mcg – 800 mcg)
- **Frequency:** Daily
- **Cycle:** 8–12 weeks

Advanced Protocol:

- **Dose:** 0.8 mg – 1.2 mg (800 mcg – 1,200 mcg)
- **Frequency:** Daily
- **Cycle:** 12 weeks

Supplements & diet: It works best with a diet high in protein and lower in sugar/carbs. Drinking lots of water is essential.

Cautions: Very mild. Some people report a little redness where they injected or a slight tummy ache, but it is generally considered much safer than full Growth Hormone.

ARA-290 (Cibinetide)

What it is / research focus: ARA-290 is a man-made copy of a small section of Erythropoietin (EPO). You might know EPO as the drug cyclists use to make more red blood cells and have endless energy.

- **The Difference:** Scientists realized that EPO does two things: 1) Makes thick blood, and 2) Heals nerves. They cut off the "blood" part and kept only the "nerve healing" part to create ARA-290.

How it works (The Master Key Story): Think of the original EPO hormone as a **Master Key** that opens two doors.

- **Door #1 (The Blood Factory):** Opens the bone marrow to make red blood cells.
- **Door #2 (The Repair Shop):** Opens the "Innate Repair Receptor" (IRR) to fix damaged nerves and lower inflammation.
- **The Hack:** ARA-290 is a modified key that **only fits Door #2**. It opens the Repair Shop to fix the nerves, but it cannot open the Blood Factory. This means you get the healing without the dangerous thick blood.

The "Strange" Fact (The Ghost Receptor): ARA-290 targets a receptor (IRR) that is essentially a "Ghost."

- **The Secret:** In a healthy person, this receptor *doesn't exist*. It only pops up on the surface of your cells when they are injured or stressed. This means ARA-290 floats harmlessly past healthy cells and *only* binds to the cells that are screaming for help. It is a "smart bomb" for inflammation.

In simple terms: It is "Nerve Fertilizer." It helps small, damaged nerves grow back and stops the burning pain caused by inflammation.

Anecdotal:

- **"The Fire Extinguisher":** This is the #1 peptide for **Small Fiber Neuropathy** (burning skin/nerves). Users often report that the constant "burning" sensation in their hands and feet starts to cool down after 2–3 weeks.
- **"The Cost Factor":** Because it has a very short lifespan in the body (half-life), clinical trials used huge doses (4 mg daily). Because this is very expensive, most researchers use smaller doses (listed below) or take it only on days when pain is bad.

Beginner Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg).
- **Frequency:** 3 times per week (SubQ).
- **Cycle:** 4–6 weeks.

Intermediate Protocol:

- **Dose:** 1.0 mg – 2.0 mg.

- **Frequency:** 3 times per week.
- **Cycle:** 6–8 weeks.

Advanced Protocol:

- **Dose:** 2.0 mg – 3.0 mg.
- **Frequency:** 3 times per week.
- **Note:** Even at this "Advanced" level, this dose is lower than what was used in successful medical trials (4 mg *daily*). This peptide has a very high safety ceiling.

Supplements & diet:

- **Omega-3s (Fish Oil):** Think of your nerves like electrical wires. The insulation around them is made of fat. You need Omega-3s to rebuild that insulation.
- **ALA (Alpha Lipoic Acid):** Often stacked with ARA-290 to help nerve pain from the inside.

Cautions:

- **Headaches:** A dull headache is the most common complaint, usually because it increases blood flow to the brain.
- **Fatigue:** Some users feel tired after the shot as the body switches into "repair mode."

B7-33

What it is / research focus: Analog of relaxin-2 studied for anti-fibrotic and cardioprotective pathways in preclinical models. It is a biased agonist at **RXFP1**, favoring ERK-dependent signaling while reducing cAMP-linked effects compared to native relaxation. Studies explore benefits in cardiac and hepatic fibrosis models.

In simple terms: A relaxin-like peptide studied to reduce scarring and protect the heart.

Anecdotal: Limited. Some users cite joint or soft-tissue comfort and easier breathing during light cardio, though experiences vary.

Research Protocol:

Beginner: 0.2–0.5 mg (200–500 mcg) SubQ 3×/week · **Cycle:** 4–6 weeks.

Intermediate: 0.5–1.0 mg SubQ 3×/week · **Cycle:** 6–8 weeks.

Advanced: 1.0–1.5 mg SubQ 3×/week · **Cycle:** 8 weeks.

Supplements & diet: General cardiometabolic support (DHA/EPA, potassium-rich foods).

Cautions: Hypotension-like feelings reported anecdotally.

BPC-157

What it is / research focus: BPC stands for "Body Protection Compound." It is a synthetic version of a peptide found in your own stomach juice.

- **The Goal:** Its natural job is to protect your stomach from its own acid. Scientists realized that if it can heal the harsh environment of the stomach, it might be able to heal *anything*—muscles, tendons, and even the brain.

How it works (The Construction Foreman Story): Think of a damaged tendon or muscle like a collapsed bridge.

- **The Workers:** Your body has plenty of workers (nutrients, blood cells), but they don't know where to go.
- **The Foreman:** BPC-157 acts like the **Foreman**. It stands at the injury site and directs traffic.
- **The Action:** It specifically tells the body to build new "roads" (blood vessels/angiogenesis) to the injury. Without these roads, the trucks carrying cement (collagen) can't get to the site to fix the bridge.

The "Strange" Fact (The Alcohol Trick): This peptide is famous for being "Anti-Intoxication."

- **The Study:** In animal studies, BPC-157 was able to block the toxic effects of alcohol and NSAIDs (like Ibuprofen/Tylenol). It protected the liver and brain so well that the subjects didn't get "drunk" or suffer liver damage, even with high doses of toxins. *Note: Do not try this at home.*

In simple terms: It is a "Healing Coordinator." It builds new blood vessels to speed up repair for injuries that usually take months to fix.

Anecdotal:

- **"The Wolverine":** This is the most popular peptide in the world for injuries. Reports of healing old tendonitis or tears in half the normal time are common.
- **"The Anhedonia" (Warning):** A small number of researchers on Reddit report a strange side effect: "Anhedonia" (inability to feel pleasure). Because BPC-157 affects the brain's dopamine system, some users feel "flat" or emotionless for a while. This usually goes away after stopping.

Beginner Protocol:

- **Dose:** 0.1 mg – 0.25 mg (100 mcg – 250 mcg).
- **Frequency:** Daily (SubQ).
- **Cycle:** 2–4 weeks.

Intermediate Protocol:

- **Dose:** 0.25 mg – 0.5 mg (250 mcg – 500 mcg).
- **Frequency:** Daily.
- **Note:** Many researchers split this into two doses (Morning and Night) because it leaves the body quickly (short half-life).

Advanced Protocol:

- **Dose:** 0.5 mg – 0.75 mg (500 mcg – 750 mcg).
- **Frequency:** Daily.
- **Cycle:** 6–8 weeks.
- **Oral vs. Injectable:** For gut issues (IBS/stomach pain), researchers usually drink it (Oral). For muscle/tendon injuries, they inject it near the pain (SubQ).

Supplements & diet:

- **Collagen & Vitamin C:** BPC-157 is the Foreman, but Collagen is the cement. You must provide the raw materials, or the Foreman has nothing to build with.

Cautions:

- **WADA Ban:** This peptide is **Banned** by the World Anti-Doping Agency (WADA). Athletes cannot use it.
- **Redness:** It can cause a red spot at the injection site.

Bronchogen (bioregulator)

What it is / research focus: Bioregulator peptide complex reported for respiratory tissue support in Russian literature. Typically described as low-molecular-weight peptides derived from organ extracts, with putative regulation of epithelial repair.

In simple terms: A mix of tiny proteins said to support the lungs.

Anecdotal: Interest around airway comfort and recovery sensations.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg SubQ weekly · **Cycle:** 6–8 weeks.

Advanced: 30–40 mg SubQ weekly · **Cycle:** 8 weeks.

Supplements & diet: Hydration, antioxidant-rich produce.

Cautions: Limited English-language primary data; monitor for idiosyncratic reactions.

Cagrilintide

What it is / research focus: Amylin analog investigated for appetite and weight-related outcomes (often in combination with GLP-1 agonists). It engages amylin receptors to slow gastric emptying and enhance satiety signals; clinical programs examine synergy with semaglutide.

In simple terms: A fullness-signal drug that can make you feel satisfied with less food.

Anecdotal: Used in appetite management; nausea common early.

Research Protocol:

Beginner: 0.1–0.3 mg SubQ weekly · **Cycle:** 8–12 weeks.

Intermediate: 0.3–0.6 mg SubQ weekly · **Cycle:** 12–16 weeks.

Advanced: 0.6–1.0 mg SubQ weekly · **Cycle:** 16 weeks.

Supplements & diet: Protein targets, fiber 25–40 g/day.

Cautions: Nausea, fullness early in cycle.

Cardiogen (bioregulator)

What it is / research focus: Bioregulator complex described for myocardial tissue support in legacy Russian sources. Claimed mechanisms include normalization of peptide synthesis and modulation of oxidative stress pathways.

In simple terms: A peptide mix aimed at helping heart tissue work better.

Anecdotal: Perceived endurance support in light training. Occasional users note calmer resting heart rate trends during deload weeks.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg SubQ weekly · **Cycle:** 6–8 weeks.

Advanced: 30–40 mg SubQ weekly · **Cycle:** 8 weeks.

Supplements & diet: Omega-3s, potassium-rich foods; avoid excessive stimulants.

Cautions: Insufficient modern clinical data.

Cartalax (bioregulator)

What it is / research focus: Cartilage-oriented bioregulator complex from legacy Eastern European research streams. Literature suggests chondrocyte metabolic support and potential modulation of inflammatory markers.

In simple terms: A peptide mix meant to support cartilage and joints.

Anecdotal: Joint comfort reports during training blocks. Some pair with collagen + vitamin C pre-training to support connective tissue routines.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg SubQ weekly · **Cycle:** 6–8 weeks.

Advanced: 30–40 mg SubQ weekly · **Cycle:** 8 weeks.

Supplements & diet: Collagen + vitamin C, omega-3s.

Cautions: Limited contemporary peer-reviewed data.

CJC-1295 no DAC (Mod GRF 1-29)

What it is / research focus: This is a man-made version of the signal your brain sends to your pituitary gland to release Growth Hormone. It is designed to work fast and then disappear, exactly like your body's natural signals.

How it works (The Doorbell Story): Your pituitary gland (which releases growth hormone) works like a **Doorbell**.

- **The Natural Way:** To get growth hormone, you have to "press the button," let it ring, and then let go. This creates a "pulse."
- **The Problem with "DAC":** The *other* version of this medicine (CJC-1295 *with* DAC) is

like taping the doorbell button down so it rings forever. This annoys the body and can cause side effects.

- **The Solution (No DAC):** This version presses the button once and then runs away. It creates a perfect, natural "pulse" of growth hormone without annoying the gland.

The "Strange" Fact (The False Identity): This peptide is actually using a fake name!

- **The Truth:** Real CJC-1295 *always* has the "DAC" part attached to it. The "No DAC" version is actually a completely different peptide called **Mod GRF 1-29**.
- **Why the confusion?** A supplement company years ago labeled it wrong, and the name stuck. If you want to sound like a pro, call it "Mod GRF."

In simple terms: It is a "Quick Text Message" to your brain. It says "Release Growth Hormone Now!" and then deletes the message so your body can rest.

Anecdotal:

- **"The Rush":** The most famous sign that it is real is the **Head Rush**. About 5–10 minutes after the shot, researchers often feel a sudden wave of warmth in their face and temples. It lasts about 20 minutes and is not dangerous—it just means the signal worked.
- **"The Stack":** You will almost never see this used alone. It is the "Batman" to Ipamorelin's "Robin." They work 5x better together than apart.

Beginner Protocol:

- **Dose:** 0.1 mg (100 mcg)
- **Frequency:** 1–2 times per day (SubQ).
- **Timing:** Immediately before bed (empty stomach).
- **Cycle:** 8–12 weeks.

Intermediate Protocol:

- **Dose:** 0.1 mg – 0.2 mg (100 mcg – 200 mcg)
- **Frequency:** 2 times per day (Morning and Night).
- **Cycle:** 8–12 weeks.

Advanced Protocol:

- **Dose:** 0.2 mg – 0.3 mg (200 mcg – 300 mcg)
- **Frequency:** 3 times per day (Morning, Post-Workout, Night).
- **Note:** Doses higher than 100mcg per shot rarely give extra benefits because the "doorbell" can only ring so loud. Frequency (ringing it more often) is better than a bigger dose.

Supplements & diet:

- **The "Carb Rule":** This signal is very fragile. If you have eaten carbs (sugar/bread) in the last hour, the insulin in your blood will block the signal. You **must** be on an empty stomach.
- **Magnesium:** Helps with the sleep benefits.

Cautions:

- **Flushing:** As mentioned, the warm face is normal.
- **Water Retention:** Much less common than with the "DAC" version, but still possible if the dose is too high.

Chonluten (bioregulator)

What it is / research focus: Bioregulator complex for respiratory/bronchial mucosa support in legacy literature. Typically presented as short peptides aimed at epithelial integrity and mucociliary function.

In simple terms: A peptide mix said to help the lining of your airways.

Anecdotal: Breathing comfort in light activity. Some note **benefits** during seasonal triggers when paired with hydration and sleep.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg SubQ weekly · **Cycle:** 6–8 weeks.

Advanced: 30–40 mg SubQ weekly · **Cycle:** 8 weeks.

Supplements & diet: Hydration; antioxidant foods.

Cautions: Sparse modern human data.

Cortagen (bioregulator)

What it is / research focus: Bioregulator complex with neuronal support claims in historical sources. Proposed actions include modulation of neurotrophic factors and synaptic plasticity markers.

In simple terms: A peptide mix aimed at helping the brain handle stress and learning.

Anecdotal: Focus/ alertness reports; sensitive users note sleep changes. Timing earlier in the day is a common practice to reduce insomnia.

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg SubQ weekly · **Cycle:** 6–8 weeks.

Advanced: 30–40 mg SubQ weekly · **Cycle:** 8 weeks.

Supplements & diet: Choline-rich foods, magnesium glycinate.

Cautions: Monitor sleep quality.

DSIP

What it is / research focus: DSIP is a famous "old school" peptide discovered in 1974. It is unique because it helps the body enter "Delta Sleep" (Deep Sleep), which is the stage where the body heals itself and releases growth hormone.

How it works (The Architect Story): Most sleeping pills act like a **Sledgehammer**—they just knock the brain out.

- **The Problem:** When you are knocked out, you don't always get good *quality* sleep. You just lose consciousness.
- **The Solution:** DSIP acts like a **Sleep Architect**. It organizes the sleep stages. It tells the brain, "Okay, now it is time for deep sleep," and helps you stay there longer naturally.

The "Strange" Fact (The Vampire Rabbits): This peptide has one of the weirdest origin stories in science.

- **The Discovery:** Scientists in the 1970s put rabbits to sleep and then filtered their blood to see if there was a "magic ingredient" causing the sleep. They found DSIP.
- **The Secret:** It wasn't just for sleep. In later studies, this same peptide was used to help people stop taking strong pain medicines (opioids) without getting sick. It seems to "reset" the brain's stress levels.

In simple terms: It is a "Deep Sleep Signal." It doesn't force you to sleep; it just helps you sleep *deeper* once you close your eyes.

Anecdotal:

- **"The Hit or Miss":** This is the most controversial part. For some researchers, it is a miracle that cures insomnia. For others, it does absolutely nothing. It seems to only work

if your natural levels are low.

- **"The Heavy Head":** If the dose is too high, users often report waking up feeling "heavy" or groggy, like they can't wake up fully for an hour.
- **"The Calm":** Many users take it during the day (in small doses) just to feel relaxed and stop stressing, even if they aren't trying to sleep.

Beginner Protocol:

- **Dose:** 0.1 mg – 0.25 mg (100 mcg – 250 mcg).
- **Frequency:** Daily (SubQ).
- **Timing:** Take 1–2 hours *before* you want to go to sleep.
- **Cycle:** 2–4 weeks.

Intermediate Protocol:

- **Dose:** 0.25 mg – 0.5 mg (250 mcg – 500 mcg).
- **Frequency:** Daily.
- **Cycle:** 4–6 weeks.

Advanced Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg).
- **Frequency:** Daily.
- **Warning:** Doses this high often cause the "heavy morning" side effect.

Supplements & diet:

- **Magnesium:** This is the perfect partner for DSIP. It relaxes the body while the peptide relaxes the brain.
- **No Caffeine:** Caffeine stays in your blood for 8 hours. If you drink coffee at 4 PM, it fights the DSIP at 10 PM.

Cautions:

- **Timing:** If you take it too early in the day, it can actually make you feel *more* awake (a paradoxical reaction). It works best when the sun is down.
- **Groggy:** Do not drive or operate machinery if you feel the "morning heaviness."

Epithalon (Epitalon)

What it is / research focus: Epithalon is perhaps the most famous "anti-aging" peptide in

history. It was discovered in the 1980s by a Russian scientist named Vladimir Khavinson.

- **The Goal:** It was originally developed to help Soviet cosmonauts and military personnel avoid "premature aging" caused by stress and radiation.

How it works (The Shoelace Story): Inside your cells, your DNA is organized into twisted strings called chromosomes. At the very end of each string is a protective cap called a **Telomere**.

- **The Analogy:** Think of a telomere like the **plastic tip (aglet) on a shoelace**.
- **The Problem:** Every time your cells divide to repair your body, that plastic tip gets a little bit shorter. When the tip is gone, the shoelace frays, and the cell dies of "old age."
- **The Fix:** Epithalon tells the body to produce an enzyme called *Telomerase*, which actually **regrows** the plastic tip. It is one of the only things known to lengthen these tips.

The "Strange" Fact (The Third Eye): Epithalon works by waking up the **Pineal Gland** (often called the "Third Eye" deep in the brain).

- **The Result:** Because the pineal gland controls your body clock, this peptide can reset your sleep cycles. This is why it was used to treat elderly people whose internal clocks had "broken," helping them sleep like teenagers again.

In simple terms: It is a "Time Machine" for your cells. It repairs the protective caps on your DNA and resets your body's internal clock.

Anecdotal:

- **"The Dream State":** The most common report is **extremely vivid dreams**. Because it reboots melatonin production, researchers often report dreaming in "4K resolution."
- **"The Deep Sleep":** Many users report waking up feeling unusually refreshed, as if they slept deeper than they have in years.
- **"The Pause":** It doesn't "feel" like anything while taking it. It is a long-term play. You take it now to look younger 10 years from now.

Research Protocol (The "Russian Protocol"): *Note: Unlike other peptides, Epithalon is not taken forever. It is taken in short "courses" just twice a year.*

Beginner Protocol:

- **Dose:** 5 mg – 10 mg
- **Frequency:** Daily (SubQ).
- **Cycle:** 10–20 days. Repeat this course every 6 months.

Intermediate Protocol:

- **Dose:** 10 mg
- **Frequency:** Daily.
- **Cycle:** 20 days. Repeat every 6 months.

Advanced Protocol:

- **Dose:** 20 mg
- **Frequency:** Daily.
- **Note:** Most modern research suggests 10 mg is the "ceiling" and 20 mg offers no extra benefit.

Supplements & diet:

- **Melatonin:** Do **NOT** take melatonin supplements while using Epithalon. The whole point is to teach your pineal gland to make its own.
- **Sleep:** You must give yourself an 8-hour sleep window to let the repair happen.

Cautions:

- **Cancer Questions:** While Russian studies show it *prevents* tumors, some Western scientists worry that "immortal cells" (long telomeres) could theoretically protect cancer cells, too.
- **No "Buzz":** Do not expect to feel energy or fat loss. This is silent cellular repair.

Follistatin-315

What it is / research focus: Myostatin-binding protein fragment explored in muscle biology; data in humans are sparse. It may interact with activin A and related TGF- β superfamily ligands, altering muscle growth signaling in models.

In simple terms: A protein piece that blocks myostatin, the "brake" on muscle growth.

Anecdotal: Strength/size pursuits; localized soreness noted. Some users report connective-tissue stiffness and prefer shorter cycles with emphasis on mobility work.

Research Protocol:

Beginner: 0.5–1.0 mg SubQ daily · **Cycle:** 10–20 days.
Intermediate: 1.0–1.5 mg SubQ daily · **Cycle:** 10–20 days.
Advanced: 1.5–2.0 mg SubQ daily · **Cycle:** 10–20 days.

Supplements & diet: High-protein diet; creatine; sleep.

Cautions: Potential for tendon stiffness anecdotally; unknown long-term effects.

Follistatin-344

What it is / research focus: Longer isoform of follistatin studied for myostatin interactions. Preclinical work indicates broad binding within the activin/myostatin axis with potential systemic effects.

In simple terms: A longer version of the muscle-growth “brake” blocker.

Anecdotal: Similar aims to 315; more injection-site sensitivity noted by some. Reports emphasize careful warm-ups and tendon care during heavy strength phases.

Research Protocol:

Beginner: 0.5–1.0 mg SubQ daily · **Cycle:** 10–20 days.

Intermediate: 1.0–1.5 mg SubQ daily · **Cycle:** 10–20 days.

Advanced: 1.5–2.0 mg SubQ daily · **Cycle:** 10–20 days.

Supplements & diet: Protein 1.6–2.2 g/kg; collagen support if heavy lifting.

Cautions: Potential for tendon stiffness anecdotally; unknown long-term effects.

FOXO4-DRI (Proxofim)

What it is / research focus: D-retro-inverso peptide antagonist of FOXO4-p53 interaction studied for senescent cell apoptosis in preclinical work. Early research suggests selective targeting of senescent cells to restore tissue homeostasis.

In simple terms: A designer peptide that tells worn-out cells to shut down.

Anecdotal: Interest in vitality; limited accessibility and tolerance data. Users often describe pulsed, short cycles with long breaks and careful monitoring.

Research Protocol:

Beginner: 2–5 mg SubQ weekly · **Cycle:** 2–4 weeks.

Intermediate: 5–10 mg SubQ weekly · **Cycle:** 4–6 weeks.

Advanced: 10–15 mg SubQ weekly · **Cycle:** 6 weeks.

Supplements & diet: Antioxidant-rich diet; hydration.

Cautions: Preclinical and anecdotal; proceed cautiously.

GHK Basic

What it is / research focus: Tripeptide (gly-his-lys) studied for wound healing and skin remodeling; often used topically; injectable research uses exist. Proposed actions include modulation of metalloproteinases, collagen synthesis, and anti-inflammatory signaling. **In simple terms:** A tiny peptide studied for helping skin repair and look healthier. **Anecdotal:** Tissue quality/cosmetic focus. Users sometimes report improved skin texture and healing appearance over multi-week blocks.

Research Protocol:

Beginner: 2–5 mg SubQ 2–3×/week · **Cycle:** 4–8 weeks.

Intermediate: 5–10 mg 2–3×/week · **Cycle:** 6–8 weeks.

Advanced: 10–15 mg 2–3×/week · **Cycle:** 8 weeks.

Supplements & diet: Vitamin C, protein sufficiency.

Cautions: Injection-site irritation possible.

GHK-Cu

What it is / research focus: GHK-Cu is a natural copper complex found in your own blood. When you are 20 years old, your blood is full of it. By age 60, levels drop by 60%.

- **The Goal:** Scientists use it to "remind" older cells how to behave like young cells. It is famous for tightening skin, healing wounds, and fixing damaged DNA.

How it works (The Renovation Crew Story): Imagine your skin and organs are an old house.

- **The Signal:** GHK-Cu acts like the **General Contractor**. It walks into the house and shouts orders.
- **The Orders:** It tells the workers to tear out the old, damaged drywall (scar tissue) and replace it with brand new, flexible drywall (collagen).
- **The Result:** It doesn't just "patch" the damage; it actually remodels the structure to look new again.

The "Strange" Fact (The Blue Blood): This peptide is naturally a **Deep Sapphire Blue** color.

- **The "Vampire" Effect:** Because it helps grow new blood vessels (angiogenesis), it is

often used in "Vampire Facials" to help the face heal instantly after microneedling.

In simple terms: It is a "Youth Signal." It tricks your body into thinking it is young again so it starts producing fresh skin and hair.

Anecdotal:

- **"The Burn":** This shot is famous for stinging. It feels like a bee sting for about 30 seconds.
- **"The Glow":** Users often report people asking if they "had work done" after 4–6 weeks because their skin looks tighter.
- **The "Copper Uglies": (Important)** If you take too much, too fast, your skin can actually get *loose* and saggy. This happens because the "Renovation Crew" tears down the old skin faster than it can build the new skin. *Less is more with this peptide.*

Beginner Protocol:

- **Dose:** 1 mg – 2 mg
- **Frequency:** 2–3 times per week (SubQ).
- **Cycle:** 4–8 weeks.

Intermediate Protocol:

- **Dose:** 2 mg – 4 mg
- **Frequency:** 2–3 times per week.
- **Cycle:** 6–8 weeks.

Advanced Protocol:

- **Dose:** 4 mg – 6 mg
- **Frequency:** 2–3 times per week.
- **Cycle:** 8 weeks max.

Supplements & diet:

- **Zinc:** This is mandatory. Copper pushes Zinc out of your body. You **must** take a Zinc supplement (separated by a few hours) to keep the balance, or you might feel sick.
- **No Vitamin C:** Do not take Vitamin C at the same time as the shot; it can break the peptide.

Cautions:

- **Toxicity:** If you start tasting pennies in your mouth, you are overdosing. Stop immediately.
- **Pain:** Injecting into fatty areas (like the butt) hurts less than the stomach.

GHRH

What it is / research focus: Native growth hormone–releasing hormone; research analogs often used due to stability. Pulsatile secretion patterns and sleep linkage are key features of physiologic GH release.

In simple terms: The body's own signal that tells the pituitary to make growth hormone.

Anecdotal: Used similarly to Mod GRF. Some report subtle body composition improvements when combined with resistance training and sleep discipline.

Research Protocol:

Beginner: 0.1 mg (100 mcg) SubQ 1–2×/day · **Cycle:** 8–12 weeks.

Intermediate: 0.1–0.2 mg 2×/day · **Cycle:** 8–12 weeks.

Advanced: 0.2–0.3 mg 2–3×/day · **Cycle:** 12 weeks.

Supplements & diet: Zinc, magnesium, sleep.

Cautions: As with other GHRH analogs.

GHRP-2

What it is / research focus: GH secretagogue; stimulates GH release via ghrelin receptor. Distinct pharmacokinetics compared to GHRP-6 with potentially less orexigenic drive.

In simple terms: A peptide that nudges the body to release growth hormone and may boost appetite.

Anecdotal: Appetite increases; tingling/flushing sometimes. Perceived synergy with Mod GRF when timed away from carbohydrate-heavy meals.

Research Protocol:

Beginner: 0.1 mg (100 mcg) SubQ 1–2×/day · **Cycle:** 8–12 weeks.

Intermediate: 0.2 mg (200 mcg) 2×/day · **Cycle:** 8–12 weeks.

Advanced: 0.3 mg (300 mcg) 2–3×/day · **Cycle:** 12 weeks.

Supplements & diet: Time doses away from large carb/fat meals.

Cautions: Appetite spikes; water retention in some.

GHRP-6

What it is / research focus: GH secretagogue; ghrelin receptor agonist with strong orexigenic effects. Often contrasted with hexarelin and ipamorelin regarding appetite and cortisol/prolactin spillover.

In simple terms: Similar to GHRP-2 but often causes a bigger increase in hunger. **Anecdotal:** Marked appetite increases common. Some report useful hunger for bulking phases; others find adherence harder during cuts.

Beginner: 0.1 mg (100 mcg) SubQ 1–2×/day · **Cycle:** 8–12 weeks.

Intermediate: 0.2 mg (200 mcg) 2×/day · **Cycle:** 8–12 weeks.

Advanced: 0.3 mg (300 mcg) 2–3×/day · **Cycle:** 12 weeks.

Supplements & diet: Plan meals intentionally to manage appetite.

Cautions: Appetite surge; potential water retention.

Glutathione

What it is / research focus: Glutathione is the "Master Antioxidant" of the human body. It is found in every single cell, but the highest concentrations are in the liver.

- **The Job:** It is the body's primary trash collector. It is responsible for hunting down heavy metals (like mercury), free radicals, and toxins, neutralizing them, and flushing them out.

How it works (The Sticky Flypaper Story): Glutathione is made of three amino acids, but the secret weapon is **Sulfur**.

- **The Trap:** Sulfur is a sticky, smelly molecule. In the body, it acts exactly like **Flypaper**.
- **The Action:** Toxins, heavy metals, and free radicals get "stuck" to the sulfur. Once they are stuck, they are neutralized and can be safely peed out.
- **The Trade-off:** This is why Glutathione smells like rotten eggs—that sulfur smell is the proof that it is potent and ready to work.

The "Strange" Fact (The ER Secret): If you go to the Emergency Room for a Tylenol overdose, the doctors don't give you a fancy drug. They give you medicine (NAC) to massively boost your **Glutathione**. Why? Because Glutathione is the *only* thing strong enough to save your liver from that level of toxicity.

In simple terms: It is the "Janitor" of your body. It cleans up the mess that stress, diet, and

pollution leave behind.

Anecdotal:

- **"The Glow":** The most famous side effect is cosmetic. High-dose users almost always report their skin becomes brighter, "glowy," or lighter in tone. This is because Glutathione switches the skin from making dark pigment (eumelanin) to light pigment (pheomelanin).
- **"The Taste":** Immediately after an injection, many researchers taste pennies or garlic in the back of their throat. This is normal—it means it is circulating in the blood.
- **The Hangover Hack:** It is legendary in biohacker circles for preventing hangovers if taken before (or after) drinking, as it helps the liver process alcohol.

Beginner Protocol (General Health):

- **Dose:** 200 mg – 400 mg
- **Frequency:** 1–2 times per week (IM).
- **Cycle:** 4–8 weeks.

Intermediate Protocol (Detox/Recovery):

- **Dose:** 400 mg – 600 mg
- **Frequency:** 2 times per week.
- **Cycle:** 6–12 weeks.

Advanced Protocol (Cosmetic/Skin Focus):

- **Dose:** 600 mg – 1,200 mg
- **Frequency:** 2–3 times per week.
- **Note:** Doses this high are usually split into multiple shots to avoid muscle pain, as the liquid is thick.

Supplements & diet:

- **Vitamin C:** This is the "Best Friend" of Glutathione. It keeps Glutathione active in the body longer.
- **NAC:** Helps the body make its own Glutathione between injections.

Cautions:

- **Asthma Warning:** This is critical. People with asthma may have a reaction to the sulfites in the solution. Start with a tiny "test dose."
- **Zinc Depletion:** Long-term use can lower zinc levels, so taking a multivitamin is smart.

Gonadorelin (GnRH)

What it is / research focus: Synthetic GnRH used in pituitary axis testing and reproductive research. Short half-life requires specific timing for axis assessments.

In simple terms: The brain's starter signal for sex hormones; mainly used to test the system.

Anecdotal: Occasionally used to probe LH/FSH responsiveness. Users emphasize conservative, test-style single administrations rather than chronic use.

Research Protocol:

Beginner: 50–100 mcg SubQ **single dose** · **Cycle:** per protocol.

Intermediate: 100–200 mcg SubQ **single dose**.

Advanced: 200–400 mcg SubQ **single dose**.

Supplements & diet: General micronutrient sufficiency.

Cautions: Endocrine axis perturbations possible; research-only use.

hGH Fragment 176-191

What it is / research focus: See AOD-9604 above; overlapping literature nomenclature.

Mechanistic focus centers on lipolysis pathways separate from full-length GH.

In simple terms: A sliver of growth hormone studied mainly for fat-loss signaling.

Anecdotal: As per AOD-9604. Users typically pair with calorie control and walking or incline cardio.

Research Protocol:

Beginner: 0.25–0.5 mg (250–500 mcg) SubQ daily · **Cycle:** 8–12 weeks.

Intermediate: 0.5–0.8 mg daily · **Cycle:** 8–12 weeks.

Advanced: 0.8–1.2 mg daily · **Cycle:** 12 weeks.

Supplements & diet: Protein-forward diet; fiber.

Cautions: GI discomfort anecdotally.

Hexarelin

What it is / research focus: Potent GHS (ghrelin receptor agonist) with strong GH-releasing effect. Some datasets suggest desensitization with continuous high-frequency dosing.

In simple terms: A strong growth-hormone releaser that can build tolerance if used too often.

Anecdotal: Joint comfort noted by some; appetite changes. Shorter cycles and lower frequencies are commonly discussed to mitigate tolerance.

Beginner: 0.05–0.1 mg (50–100 mcg) SubQ 1–2×/day · **Cycle:** 4–8 weeks.

Intermediate: 0.1–0.2 mg 2×/day · **Cycle:** 6–8 weeks.

Advanced: 0.2–0.3 mg 2–3×/day · **Cycle:** 8–12 weeks.

Supplements & diet: As with other GHS peptides.

Cautions: Desensitization risk discussed anecdotally.

Humanin

What it is / research focus: Mitochondria-derived peptide studied for cytoprotective signaling. Investigations include apoptosis resistance, insulin sensitivity, and neuroprotective pathways in models.

In simple terms: A tiny protein from mitochondria studied for protecting cells from stress.

Anecdotal: Wellness/energy focus. Some note improved perceived resilience to intense training weeks with consistent dosing.

Research Protocol:

Beginner: 2–5 mg SubQ 2×/week · **Cycle:** 4–8 weeks.

Intermediate: 5–10 mg 2×/week · **Cycle:** 6–8 weeks.

Advanced: 10–15 mg 2×/week · **Cycle:** 8–12 weeks.

Supplements & diet: Polyphenol-rich diet; exercise.

Cautions: Limited human trials.

IGF-1 DES

What it is / research focus: Short-acting IGF-1 analog investigated for local tissue signaling. Its rapid clearance favors peri-site microdosing strategies in research contexts.

In simple terms: A very short-acting IGF-1 used right where you want the effect.

Anecdotal: Target-site micro-dosing around training. Users emphasize careful carbohydrate intake to avoid hypoglycemic symptoms.

Research Protocol:

Beginner: 10–20 mcg SubQ per site 1×/day · **Cycle:** 4–6 weeks.

Intermediate: 20–40 mcg per site 1–2×/day · **Cycle:** 4–6 weeks.

Advanced: 40–60 mcg per site 1–2×/day · **Cycle:** 6 weeks.

Supplements & diet: Protein sufficiency; carb timing if training.

Cautions: Hypoglycemia potential; research-only.

IGF-1 LR3

What it is / research focus: Long-acting IGF-1 analog studied for anabolic/insulin-like pathways. Extended half-life supports once-daily systemic protocols in research settings.

In simple terms: A long-lasting form of IGF-1 that circulates through the body. **Anecdotal:** Systemic dosing patterns vary widely. Appetite swings and pump sensation are frequently mentioned; hypoglycemia risk is a recurring caution.

Research Protocol:

Beginner: 20–40 mcg SubQ daily · **Cycle:** 3–4 weeks.

Intermediate: 40–60 mcg SubQ daily · **Cycle:** 4–6 weeks.

Advanced: 60–100 mcg SubQ daily · **Cycle:** 4–6 weeks.

Supplements & diet: Carbohydrate awareness protein targets.

Cautions: Hypoglycemia risk; proceed carefully.

Ipamorelin

What it is / research focus: Selective GH secretagogue with minimal prolactin/cortisol spillover in studies. Often chosen for gentle GH pulse support with fewer side effects than older GHRPs.

In simple terms: A gentler growth-hormone releaser with fewer side effects.

Anecdotal: Commonly paired with Mod GRF for synergy. Users often dose pre-sleep or upon waking to align with fasting windows.

Research Protocol:

Beginner: 0.1 mg (100 mcg) SubQ 1–2×/day · **Cycle:** 8–12 weeks.

Intermediate: 0.2 mg (200 mcg) 2×/day · **Cycle:** 8–12 weeks.

Advanced: 0.3 mg (300 mcg) 2–3×/day · **Cycle:** 12 weeks.

Supplements & diet: Sleep hygiene; zinc/magnesium; carb-free window around injections.

Cautions: Headache, flushing uncommon.

Kisspeptin-10

What it is / research focus: Decapeptide regulating GnRH release; used in reproductive axis research. Short-term tests can assess hypothalamic-pituitary responsiveness.

In simple terms: A hormone fragment that starts the chain that leads to sex hormones.

Anecdotal: Limited; sometimes used for axis testing. Conservative single-dose approaches are most common in logs.

Research Protocol:

Beginner: 50–100 mcg SubQ single dose · **Cycle:** per protocol.

Intermediate: 100–200 mcg SubQ single dose.

Advanced: 200–400 mcg SubQ single dose.

Supplements & diet: General.

KPV (Ac-KPV-NH2)

What it is / research focus: KPV is a tiny, 3-amino acid fragment taken from the larger hormone alpha-MSH.

- **The Goal:** Scientists wanted the incredible anti-inflammatory power of alpha-MSH without the side effects (like darkening skin or causing erections). KPV is that isolated "healing" signal.

How it works (The Fire Extinguisher Story): Imagine inflammation in your body is a kitchen fire.

- **The "Master Switch" (NF-κB):** In your cells, there is a switch called NF-κB. When it gets flipped, it sprays "gasoline" (inflammation) everywhere to fight infection.
- **The Solution:** KPV acts like a **Fire Extinguisher**. It walks into the cell and physically holds the switch down so it cannot flip.
- **The Result:** It stops the inflammation at the very start of the process, cooling down the gut, skin, and joints.

The "Strange" Fact (The Mold Killer): Most peptides just signal cells to "grow" or "heal." KPV is different because it is also **Antimicrobial**.

- **The Surprise:** Research shows it can actually kill dangerous microbes like *Staph* bacteria and *Candida* (yeast) on contact. This makes it a top choice for researchers studying "Mold Toxicity" or Lyme disease, where the body is fighting a chronic invader.

In simple terms: It is a "Cooling Agent." It turns off the inflammation switch in your cells and fights off bad bacteria/fungus at the same time.

Anecdotal:

- **Oral vs. Injectable:** Because KPV is so tiny, it is one of the few peptides that survives digestion. Many researchers prefer **Oral Capsules** for gut issues (Crohn's/IBS) because the pill goes straight to the inflamed stomach lining.
- **"The Psoriasis Fix":** Users often report that stubborn skin red patches fade within weeks. It is often mixed into creams for this reason.
- **"The Calm":** Unlike steroids that make you jittery, KPV is reported to have a subtle calming effect on the mood.

Beginner Protocol:

- **Dose:** 0.1 mg – 0.2 mg (100 mcg – 200 mcg).
- **Frequency:** Daily (SubQ).
- **Cycle:** 4–8 weeks.

Intermediate Protocol:

- **Dose:** 0.2 mg – 0.4 mg (200 mcg – 400 mcg).
- **Frequency:** Daily.
- **Cycle:** 6–8 weeks.

Advanced Protocol:

- **Dose:** 0.4 mg – 0.6 mg (400 mcg – 600 mcg).
- **Frequency:** Daily.
- **Note:** For severe autoimmune flares (like Lyme or Colitis), researchers often split this into two doses (morning and night).

Supplements & diet:

- **Omega-3s:** KPV stops the fire; Omega-3s clean up the smoke damage. They work perfectly together.
- **Curcumin:** Another natural NF-κB blocker that stacks well with KPV.

Cautions:

- **Very Safe:** Because it is a natural fragment found in the body, side effects are extremely rare.
- **Injection Pain:** It can sting slightly, but much less than other peptides.

Liraglutide

What it is / research focus: Liraglutide is the "older brother" to Semaglutide. It was one of the first successful daily injections for weight loss and diabetes.

- **The Goal:** It mimics the GLP-1 hormone to control blood sugar and appetite, but because it is an older version, it doesn't last as long in the body, so it must be taken every day.

How it works (The Traffic Cop Story): Think of your stomach like a busy highway tunnel. Usually, food zooms through it at 60 mph.

- **The Problem:** When food zooms through fast, you get hungry again quickly.

- **The Traffic Cop:** Liraglutide stands at the entrance of the tunnel holding a **Stop Sign**.
- **The Result:** It forces the food to slow down to 5 mph. Because the "tunnel" (stomach) stays full of cars (food) for hours, your brain thinks, "I am stuffed," and you stop eating.

The "Strange" Fact (The Anglerfish Origin): Before scientists found similar hormones in Gila Monsters, the original research for GLP-1 actually started with **Anglerfish!**

- **The Discovery:** In the 1980s, scientists went fishing for Anglerfish because they have a special organ that produces massive amounts of hormones. This research helped identify the very first GLP-1 signals that eventually led to the invention of Liraglutide.

In simple terms: It is a "Daily Diet Shot." It slows down your digestion so you feel full all day, but you have to take it every morning.

Anecdotal:

- **"The Needle Fatigue":** Because this requires a shot **every single day**, many researchers eventually switch to Semaglutide (weekly) just to avoid the daily poke.
- **"The Gentle Start":** Some users report that Liraglutide feels "gentler" or easier to control than Semaglutide. Because it leaves the body faster (13 hours vs. 7 days), if you get sick/nauseous, you can just stop taking it and feel better the next day. With Semaglutide, you are stuck with the sickness for a week.
- **"The Alcohol Eraser":** Like other GLP-1s, researchers often report losing the taste for alcohol entirely.

Beginner Protocol (The Staircase):

- *Note: You must follow the "Staircase" to avoid getting sick. Do not jump ahead.*
- **Dose:** 0.6 mg
- **Frequency:** Daily (SubQ)
- **Cycle:** Week 1

Intermediate Protocol:

- **Dose:** 1.2 mg – 1.8 mg
- **Frequency:** Daily
- **Cycle:** Increase by 0.6 mg each week. (Week 2 = 1.2 mg, Week 3 = 1.8 mg).
- **Note:** Many researchers stop at 1.8 mg (the Victoza dose) if they are still losing weight.

Advanced Protocol:

- **Dose:** 2.4 mg – 3.0 mg

- **Frequency:** Daily
- **Note:** 3.0 mg is the maximum dose (Saxenda dose). Going higher does not help and only increases nausea.

Supplements & diet:

- **Fiber:** This is mandatory. Since the "Traffic Cop" slows down your gut, things can get backed up (constipation). You need fiber to keep traffic moving.
- **Protein Shakes:** You might not feel like chewing food. Drinking your protein ensures you don't lose muscle.

Cautions:

- **Thyroid Tumors:** Like all drugs in this family, it caused thyroid tumors in rats. People with a family history of Medullary Thyroid Cancer should not use it.
- **Pancreatitis:** If you feel a sharp pain in your stomach that goes through to your back, stop immediately.

Livagen (bioregulator)

What it is / research focus: Bioregulator complex associated with liver tissue in regional literature. Proposed benefits include support of hepatocyte peptide synthesis and antioxidative defenses.

In simple terms: A peptide mix meant to support the liver's normal work.

Anecdotal: Digestive comfort reports. Users sometimes combine with reduced alcohol intake and a nutrient-dense diet.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly · **Cycle:** 6–8 weeks.

Advanced: 30–40 mg weekly · **Cycle:** 8 weeks.

Supplements & diet: Emphasize whole-food diet; limit alcohol.

Cautions: Sparse modern human trials.

LL-37 (CAP-18)

What it is / research focus: Human cathelicidin antimicrobial peptide; studied for innate immunity and wound models. Research examines biofilm disruption and immunomodulatory roles in mucosa and skin.

In simple terms: One of the body's natural germ-fighting peptides.

Anecdotal: Skin support interest; injection-site burning reported. Some users experience transient flares before improvement when used in skin-adjacent protocols.

Research Protocol:

Beginner: 0.1–0.25 mg (100–250 mcg) SubQ 2–3×/week · **Cycle:** 3–4 weeks.

Intermediate: 0.25–0.5 mg 2–3×/week · **Cycle:** 4–6 weeks.

Advanced: 0.5–1.0 mg 2–3×/week · **Cycle:** 6 weeks.

Supplements & diet: Vitamin D sufficiency; zinc status.

Cautions: Inflammation or flares reported in sensitive users.

Melanotan II (MT-2)

What it is / research focus: Melanotan II is a synthetic version of a hormone naturally produced in the brain (alpha-MSH). It was originally designed to protect fair-skinned people from skin cancer by giving them a protective tan without needing to burn in the sun first.

How it works (The Internal Sunblock): Usually, your skin only creates melanin (pigment) *after* it has been damaged by the sun. It's a defense mechanism.

- **The Hack:** MT-2 tricks your skin cells into thinking they are under attack from UV rays, even if the subject is sitting in a dark room.
- **The Result:** The body pumps out massive amounts of melanin to "build a shield." When the subject finally does go in the sun, they turn brown almost instantly instead of burning red.

The "Strange" Fact (The Barbie Drug): This peptide is famous for two weird side effects that earned it the nickname "The Barbie Drug": it makes you tan, and it suppresses your appetite so you get thin.

- **The Accident:** During early testing, a scientist accidentally injected himself with **double** the intended dose. He didn't just get a tan—he got an 8-hour erection. This accidental

discovery eventually led to the creation of **PT-141** (the sex peptide) by removing the tanning part of the molecule!

In simple terms: It is a "Sunless Tan Shot." It forces your body to produce a deep, natural tan while also crushing your appetite and boosting libido.

Anecdotal:

- **"The Tax":** Almost every new researcher pays a "tax" for the tan: **Nausea**. It hits 15 minutes after the shot and can feel like motion sickness.
- **"The Dirty Tan":** If you take too much too fast, your moles and freckles will turn black *before* your skin gets tan. This is why the modern trend is "Low and Slow."
- **"The Glow":** Users often report a flushed, red face for an hour after injecting.

Beginner Protocol (The "Low & Slow" Approach):

- **Goal:** To avoid turning moles black.
- **Dose:** 0.1 mg – 0.25 mg (100 mcg – 250 mcg).
- **Frequency:** Daily for 1–2 weeks.
- **Note:** Do not increase the dose until you have zero nausea.

Intermediate Protocol:

- **Dose:** 0.25 mg – 0.5 mg (250 mcg – 500 mcg).
- **Frequency:** Daily until desired tan is reached.
- **Maintenance:** Once the subject is tan enough, 0.5 mg **once a week** is usually enough to keep it.

Advanced Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg).
- **Warning:** Doses this high almost guaranteed severe nausea and extremely dark moles. Most modern researchers agree that 1 mg is unnecessary.

Supplements & diet:

- **Antihistamine:** Taking a Zyrtec or Claritin 30 minutes *before* the shot is a common hack to stop the nausea.
- **Ginger:** Chewing raw ginger helps the stomach settle.

Cautions:

- **Priapism:** Men may experience painful, unwanted erections that last for hours.
- **Mole Changes:** If a mole changes shape or gets irregular borders, stop immediately. MT-2 darkens moles, but it can make it hard to spot actual skin cancer.

MGF IGF-1 Ec

What it is / research focus: Mechano-growth factor splice variant of IGF-1; studied in muscle repair models. Hypothesized to act during early repair windows following mechanical overload.

In simple terms: An IGF-1 variant linked to early muscle repair after hard workouts.

Anecdotal: Localized micro-dosing post-training. Athletes often limit total exposure and emphasize technique to avoid overuse injuries.

Research Protocol:

Beginner: 50–100 mcg SubQ per site post-training · **Cycle:** 3–4 weeks.

Intermediate: 100–150 mcg per site · **Cycle:** 4–6 weeks.

Advanced: 150–200 mcg per site · **Cycle:** 4–6 weeks.

Supplements & diet: Protein + carbs post-training.

Cautions: Hypoglycemia potential with IGF-axis peptides.

MOTS-c

What it is / research focus: MOTS-c is unique because it doesn't come from your cell's main DNA; it comes from the DNA inside your mitochondria. It is often called an "exercise mimetic" because it activates the same signals (like AMPK) that lifting weights or running does.

- **The Goal:** To trick the body into burning fat and clearing sugar from the blood, even without a workout.

How it works (The "Backup Generator" Story): Your body has two ways to make energy: the main power grid (food) and the backup generators (mitochondria).

- **The Signal:** When you exercise hard, your mitochondria send a text message to the rest of the body saying, "We are working hard, send more fuel!"
- **The Hack:** MOTS-c is that text message. Injecting it tells your muscles to open their doors for sugar and burn fat, even if the subject is sitting on the couch.

The "Strange" Fact (The Alien DNA): This peptide wasn't even discovered until 2015 because scientists were looking in the wrong place. It is encoded in the **Mitochondrial DNA**, not the

Human Nucleus. Since mitochondria were originally ancient bacteria that started living inside humans, you could say this is a signal from our "inner alien" to keep us fit.

In simple terms: It is "Exercise in a Bottle." It turns on the body's metabolism heaters to burn fuel exactly like a hard workout does.

Anecdotal:

- **"The Sting":** This is the most famous part of MOTS-c. It **burns** when injected. Researchers call it "spicy water." The pain only lasts a minute, but it is surprising if you aren't ready for it.
- **The "Third Lung":** Endurance athletes (runners/cyclists) report that they simply don't get winded as easily. It feels like they have an extra lung during cardio.

Beginner Protocol:

- **Dose:** 5 mg
- **Frequency:** 2 times per week (SubQ)
- **Cycle:** 4 weeks.
- **Note:** Taking it right before a workout is the most popular timing to maximize the "energy" effect.

Intermediate Protocol:

- **Dose:** 10 mg
- **Frequency:** 2–3 times per week.
- **Cycle:** 4–6 weeks.

Advanced Protocol:

- **Dose:** 15 mg – 20 mg
- **Frequency:** 3 times per week.
- **Warning:** The "sting" gets worse as the dose gets higher. Many researchers split this into two different shots to reduce the pain.

Supplements & diet:

- **B-Vitamins (Folate):** This is critical. MOTS-c uses up your body's folate (Vitamin B9). You **must** take a B-Complex or Methyl-Folate supplement, or you might get a headache or feel "off."

Cautions:

- **Injection Pain:** As mentioned, the burning sensation is normal and does not mean it is

infected.

- **Insulin Drop:** Because it clears sugar from the blood so fast, it can cause low blood sugar (hypoglycemia) if you haven't eaten.

N-Acetyl Epithalon Amide

What it is / research focus: Epithalon analog with similar telomere/circadian-related research history; human evidence remains limited. Amidation may alter stability and tissue distribution compared with the parent tetrapeptide.

In simple terms: A tiny peptide like Epithalon, linked to body-clock and aging research.

Anecdotal: As with Epithalon—short blocks, vivid dreams reported by some.

Research Protocol:

Beginner: 5–10 mg SubQ daily · **Cycle:** 10–20 days.

Intermediate: 10–15 mg daily.

Advanced: 15–20 mg daily.

Supplements & diet: Sleep hygiene.

Cautions: Limited modern RCTs.

N-Acetyl Selank Amide

What it is / research focus: Selank derivative; published studies largely intranasal; SubQ/IM data are sparse. Interest focuses on GABAergic modulation and anxiolytic-like effects in models.

In simple terms: A calmer-mood peptide mostly tested as a nose spray; shots are less studied.

Anecdotal: Calm-focus reports; some sedation when dosed late.

Research Protocol:

Route note: SubQ/IM research data limited.

Beginner: 0.2–0.5 mg (200–500 mcg) SubQ daily · **Cycle:** 2–4 weeks.

Intermediate: 0.5–1.0 mg daily.

Advanced: 1.0–1.5 mg daily.

Supplements & diet: Magnesium, L-theanine PRN.

Cautions: Sedation in some.

N-Acetyl Semax Amide

What it is / research focus: Semax derivative; literature concentrates on intranasal routes; SubQ/IM uses are exploratory. Targets include neurotrophic factor expression and stress-response modulation.

In simple terms: A focus/learning peptide mostly tested as a spray; shots are less explored.

Anecdotal: Focus/mood notes; SubQ use is less common.

Research Protocol: Route note: SubQ/IM research data limited.

Beginner: 0.2–0.5 mg daily · **Cycle:** 2–4 weeks.

Intermediate: 0.5–1.0 mg daily.

Advanced: 1.0–1.5 mg daily.

Supplements & diet: DHA/EPA; magnesium.

Cautions: Headache possible.

NAD+

What it is / research focus: NAD+ is found in every single living cell. It is the fuel that powers the engines (mitochondria) inside your cells.

- **The "Aging" Problem:** Research shows that by age 50, most people have lost **50%** of their natural NAD+.
- **The Result:** Without it, the cells can't make energy, and they can't repair broken DNA. This is why scientists believe we get "slower" and "older."

How it works (The Spark Plug Story): Imagine your body is a sports car.

- **The Fuel:** Food is the gasoline.
- **The Problem:** You can fill the tank with gas, but if the **Spark Plugs** are dead, the car won't start.
- **The Solution:** NAD+ is the spark. It takes the food you eat and actually turns it into the electricity that keeps your heart beating and brain thinking.

In simple terms: It is a "Battery Charger" for your cells. It refills the energy tank so your body can fix itself like it did when you were younger.

The "Strange" Fact (The Addiction Killer): NAD+ is famous in rehab centers. High doses are

used to help people stop drinking alcohol or using drugs. It seems to "scrub" the crave-receptors in the brain, helping people feel normal again without withdrawals. It's like a reset button for bad habits.

Anecdotal:

- **"The Spicy Shot":** This is notorious for the "burn." Injecting NAD+ often feels like a bee sting or "spicy water" going in. This is normal.
- **"The Clarity":** Users rarely report a caffeine-like "buzz." Instead, they describe it as a "lifting of the fog"—colors seem brighter, and complex tasks feel easier.
- **"The Flush":** If injected too fast, it can cause a sudden wave of heat or a heavy feeling in the chest/stomach. This passes in minutes.

Beginner Protocol:

- **Dose:** 50 mg – 100 mg
- **Frequency:** Weekly (IM is preferred to reduce burning).
- **Cycle:** 4–6 weeks.

Intermediate Protocol:

- **Dose:** 100 mg – 250 mg
- **Frequency:** Weekly.
- **Note:** At this dose, many researchers split the shot into two smaller doses to avoid the "heavy chest" feeling.

Advanced Protocol:

- **Dose:** 250 mg – 500 mg
- **Frequency:** Weekly.
- **Warning:** 500 mg is a very high volume. This is often better handled as two separate shots or a slow IV drip.

Supplements & diet:

- **TMG (The Sidekick):** Taking NAD+ uses up your body's "methyl" groups. Adding a TMG supplement is the standard "best practice" to prevent headaches or fatigue.
- **Resveratrol:** Often stacked with NAD+ to help activate the "longevity genes" (sirtuins).

Cautions:

- **The "Stomach Cramp":** If the dose hits the blood too fast, it causes smooth muscles (like the stomach and intestines) to contract. This feels like a sudden, deep cramp.
Solution: Inject very slowly.

- **Heart Rate:** A temporary increase in heart rate is common immediately after dosing.

Oxytocin

What it is / research focus: Oxytocin is famous as the "Love Hormone" because it is released during hugging, bonding, and childbirth. However, researchers are studying it for much more than just romance. It is being investigated to help with social anxiety, autism, and even to lower cortisol (stress) levels in the body.

How it works (The Wi-Fi Story): Think of two people like two separate computers. Usually, they are disconnected.

- **The Connection:** Oxytocin acts like the **Wi-Fi Password**.
- **The Result:** When the brain has enough of it, it drops the "firewall" (fear/suspicion) and allows a strong connection to form. It makes the subject feel safe, understood, and "in sync" with the people around them.

The "Strange" Fact (The Us vs. Them Paradox): Most people think Oxytocin makes you love *everyone*. That is false.

- **The Truth:** Research suggests it only increases trust for your "Tribe" (friends and family). It can actually make you *more* suspicious or competitive toward "Outsiders" or strangers. It's not just a "Love Drug"—it's a "Loyalty Drug."

In simple terms: It is a "Liquid Hug." It lowers social fear and helps you feel close to people you trust.

Anecdotal:

- **The "Warmth":** Users often report a physical sensation of warmth in the chest within 15–20 minutes of a SubQ shot.
- **The "Glow":** It is frequently used before social events to kill "social anxiety" without the sloppiness of alcohol.
- **SubQ vs. Nasal:** While nasal sprays are common, many researchers prefer the SubQ shot because they claim the effects are more consistent and "hit" faster (usually within 10–15 minutes), whereas sprays can be hit-or-miss depending on a stuffy nose.

Beginner Protocol:

- **Dose:** 5 IU – 10 IU
- **Frequency:** As needed (PRN).

- **Timing:** 20–30 minutes before a social event or stressful situation.

Intermediate Protocol:

- **Dose:** 10 IU – 20 IU
- **Frequency:** As needed.

Advanced Protocol:

- **Dose:** 20 IU – 40 IU
- **Frequency:** As needed.
- **Note:** Doses above 20 IU can sometimes cause a "weepy" or overly emotional feeling in some researchers.

Supplements & diet:

- **Magnesium:** Helps relax the muscles to match the relaxed mind.
- **No Alcohol:** Alcohol can numb the effects of Oxytocin, making the research pointless.

Cautions:

- **The "Flush":** A sudden warm flushing of the face is very common immediately after injection.
- **Emotional Sensitivity:** Because it opens you up emotionally, do not use this around toxic people or during an argument. You may feel hurt more easily than usual.

Ovagen (bioregulator)

What it is / research focus: Bioregulator associated with liver/GI support in regional literature; claims include normalization of peptide metabolism and antioxidant defenses. **In**

simple terms: A peptide mix meant to support liver and digestion.

Anecdotal: Digestive comfort.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Whole-food diet.

Cautions: Limited modern trials.

Pancragen (bioregulator)

What it is / research focus: Bioregulator with pancreatic tissue focus; aims to support endocrine/exocrine peptide signaling balance.

In simple terms: A peptide mix aimed at helping the pancreas work normally. **Anecdotal:** Appetite normalization reports.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Balanced macros; fiber.

Cautions: As above.

PE-22-28

What it is / research focus: Peptide investigated for neurotrophic properties in early research, including BDNF-adjacent signaling hypotheses.

In simple terms: A small peptide studied for helping brain cells grow and connect.

Anecdotal: Mood/drive notes.

Research Protocol:

Beginner: 0.2–0.5 mg (200–500 mcg) SubQ daily · **Cycle:** 2–4 weeks.

Intermediate: 0.5–1.0 mg daily.

Advanced: 1.0–1.5 mg daily.

Supplements & diet: DHA/EPA, sleep hygiene.

Cautions: Limited human data.

PEG-MGF

What it is / research focus: Pegylated MGF for extended half-life; studied in muscle repair models with systemic exposure.

In simple terms: A longer-lasting muscle-repair signal used a couple of times per week.

Anecdotal: Systemic dosing post-training.

Research Protocol:

Beginner: 100–200 mcg SubQ **2×/week** · **Cycle:** 4–6 weeks.

Intermediate: 200–300 mcg 2×/week.

Advanced: 300–400 mcg 2×/week.

Supplements & diet: Protein/carbs post-training

Cautions: As with IGF-axis peptides.

Pinealon (bioregulator)

What it is / research focus: Bioregulator with CNS emphasis in legacy literature; posited to support synaptic plasticity and antioxidant defenses.

In simple terms: A peptide mix aimed at calm focus and brain health.

Anecdotal: Calm focus.

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Magnesium; sleep.

Cautions: Minimal modern trials.

PNC-27

What it is / research focus: Peptide researched for tumor-selective membrane disruption in preclinical work; binds to overexpressed receptors on certain cancer cell lines.

In simple terms: A lab peptide that targets certain cancer cells in dishes and animals.

Anecdotal: Very limited.

Research Protocol:

Beginner: 0.5–1.0 mg SubQ 2–3×/week · **Cycle:** 4–6 weeks.

Intermediate: 1.0–2.0 mg 2–3×/week.

Advanced: 2.0–3.0 mg 2–3×/week.

Supplements & diet: Antioxidants from food.

Cautions: Strictly research; safety unknown.

Prostamax (bioregulator)

What it is / research focus: Bioregulator linked with prostate tissue; proposed to support epithelial health and peptide metabolism.

In simple terms: A peptide mix meant to support normal prostate function.

Anecdotal: Urinary comfort noted.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Hydration; avoid unproven herb stacks here; stick to basics.

Cautions: Monitor tolerance.

PT-141 (Bremelanotide)

What it is / research focus: This is the only peptide on the market that is FDA-approved specifically to boost sexual desire. Unlike pills (like Viagra) that just fix "blood flow," PT-141 works directly on the brain.

The "Strange" Story (How it was found): This was originally invented as a **sunless tanning** drug (Melanotan II).

- **The Accident:** During the tanning experiments, the male researchers noticed a very unexpected side effect: they were getting spontaneous erections that lasted for hours.
- **The Pivot:** Scientists realized they had found a "switch" in the brain for arousal. They removed the tanning part of the molecule and kept the "fun" part, creating PT-141.

In simple terms: It is a "Brain Switch" for libido. It tells your brain "I am in the mood," which then tells your body to get ready.

Anecdotal:

- **The "Tax":** There is a price to pay for the fun—**Nausea**. almost every researcher reports feeling sick to their stomach for the first 20–30 minutes after the shot.

- **The Timing:** It is *not* instant. It usually takes 2 to 4 hours to kick in, so researchers have to plan ahead.
- **The "Blue Lip" Effect:** It effectively lowers blood pressure in some people, which can make lips look slightly blue or purple for a short time.

Beginner Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg)
- **Frequency:** As needed (only when you plan to use it).
- **Timing:** Take it 2–4 hours *before* the planned activity.

Intermediate Protocol:

- **Dose:** 1.0 mg – 1.75 mg
- **Frequency:** As needed.
- **Note:** 1.75 mg is the standard dose in the official medical version (Vyleesi).

Advanced Protocol:

- **Dose:** 1.75 mg – 2.0 mg
- **Frequency:** As needed.
- **Warning:** Going higher than 2 mg usually just makes the nausea worse, not the benefits better.

Supplements & diet:

- **Ginger & Dramamine:** Since nausea is almost guaranteed for new users, taking a ginger pill or motion-sickness pill 30 minutes *before* the shot is a common researcher hack.

Cautions:

- **Blood Pressure:** It can temporarily raise blood pressure after the initial drop.
- **Frequency:** Do not use this more than 8 times a month. Your brain gets used to it very fast, and it will stop working if you abuse it.

P21 (P021)

What it is / research focus: Peptide studied for neurotrophic/neuroprotective pathways in early investigations; interacts with BDNF-linked signaling.

In simple terms: A peptide being studied to help brain cells grow and protect themselves.

Anecdotal: Focus and memory discussion threads.

Research Protocol:

Beginner: 0.2–0.5 mg (200–500 mcg) SubQ daily · **Cycle:** 2–4 weeks.

Intermediate: 0.5–1.0 mg daily.

Advanced: 1.0–1.5 mg daily.

Supplements & diet: Omega-3s; sleep.

Cautions: Insufficient human data.

Retatrutide

What it is / research focus: Retatrutide is currently the most powerful metabolic drug in existence. It is nicknamed "Triple G" because it targets **three** separate hunger and metabolism signals (GLP-1, GIP, and Glucagon).

- **The Goal:** While other drugs just lower appetite, Retatrutide is designed to actively **burn** energy.

How it works (The "Three Engine" Story): Imagine your metabolism is a spaceship.

- **Engine 1 (GLP-1):** The "Brakes." It slows down digestion so you feel full. (This is what Ozempic does).
- **Engine 2 (GIP):** The "Storage Manager." It helps your body handle sugar so it doesn't get stored as fat. (This is what Mounjaro adds).
- **Engine 3 (Glucagon):** The "Afterburner." This is unique to Retatrutide. It turns up the body's thermostat, forcing it to burn calories for heat, even while you are sleeping.

The "Strange" Fact (The Glucagon Paradox): You might know that *Glucagon* normally raises blood sugar (it's the opposite of insulin). So why is it in a diabetes drug?

- **The Magic Trick:** The other two ingredients (GLP-1 and GIP) are so good at lowering sugar that they cancel out the sugar-spike from the Glucagon. This leaves *only* the fat-burning benefits behind. It's like drinking an energy drink that somehow helps you sleep.

Anecdotal:

- **"The Melt":** Researchers report weight loss happens much faster than with other compounds, often calling it "scary fast."
- **Skin Sensitivity:** A very strange but common side effect is "Allodynia"—the skin on the back, arms, or neck might feel sensitive, like a mild sunburn, even though nothing is there.

- **Heart Rate:** Because of the "Afterburner" effect, it is normal for the resting heart rate to go up by 10–15 beats per minute.

Beginner Protocol (The "Test Flight"):

- **Dose:** 1 mg – 2 mg
 - *Note:* Clinical trials start at 2 mg, but many researchers start at 1 mg to be safe.
- **Frequency:** Weekly (SubQ)
- **Cycle:** 4 weeks.

Intermediate Protocol:

- **Dose:** 4 mg – 6 mg
- **Frequency:** Weekly
- **Cycle:** 4 weeks at each dose.

Advanced Protocol (The Clinical Standard):

- **Dose:** 9 mg – 12 mg
- **Frequency:** Weekly
- **Note:** The maximum dose in clinical trials is 12 mg. Most researchers never need to go this high because the lower doses are so strong.

Supplements & diet:

- **Electrolytes:** The rapid weight loss flushes water out of the body. You must replace salt and potassium or you will feel dizzy.
- **Calorie Tracking:** The appetite suppression is so strong that you might accidentally starve yourself. You *must* eat to keep the metabolism running.

Cautions:

- **Heart Rate:** If resting heart rate stays too high (over 100 bpm while sitting), research usually stops.
- **Muscle Loss:** Because it burns energy so fast, it will burn muscle if you don't eat enough protein.

Selank

What it is / research focus: Selank is a famous anti-anxiety peptide from Russia. It was created by taking a natural peptide produced by your spleen (called *Tufts*) and adding a "tail"

to it so it could survive in the body.

- **The Goal:** Scientists wanted a medicine that could stop fear and anxiety *without* making soldiers or pilots feel sleepy or slow.

How it works (The Noise Canceling Headphones): Think of anxiety like a loud, static noise in your head that makes it hard to think.

- **Old Medicine (Valium/Xanax):** These work like "earplugs." They block the noise, but they also block *everything else*, making you feel numb or tired.
- **The Upgrade (Selank):** Selank acts like **Noise Canceling Headphones**. It actively filters out the static (stress) but lets the music (clear thinking) come through perfectly.
- **The "Bodyguard":** It also stops your body from breaking down its own natural painkillers (enkephalins), so you feel safer and calmer naturally.

In simple terms: It effectively turns down the "stress volume" in your brain so you can focus, but it doesn't make you feel like a zombie.

The "Strange" Bonus: Believe it or not, this peptide is actually an **Immune System** booster. Because it is based on a peptide from the spleen, research shows it may actually help fight off viruses (like the flu) while it calms you down. It's like a stress-reliever and a vitamin C shot in one.

Anecdotal:

- **The "Metallic Taste":** Since this is almost always taken as a **Nasal Spray**, users often report a metallic or coin-like taste in the back of their throat. This is normal and means it worked.
- **The Feeling:** Reports rarely mention a "buzz." Instead, users describe it as simply realizing they aren't worried anymore. It is often described as "smooth" or "glassy" mental clarity.

Research Protocol (Nasal vs. Shot): *Note: While most peptides are injected, Selank is famous for being a Nasal Spray. This is because it travels up the nose nerves directly into the brain.*

Beginner Protocol (Nasal Spray):

- **Dose:** 1–2 sprays in each nostril.
- **Frequency:** 2 to 3 times per day.
- **Cycle:** 2 weeks on, 1 week off.

Intermediate Protocol (SubQ Injection):

- **Dose:** 250 mcg – 300 mcg (0.25 mg – 0.3 mg).

- **Frequency:** Daily (usually morning).
- **Note:** Some researchers believe injections are stronger, but the nasal spray is the "classic" Russian method.

Advanced Protocol:

- **Dose:** 300 mcg – 1,000 mcg (1 mg).
- **Frequency:** Daily.
- **Cycle:** Up to 4 weeks.

Supplements & diet:

- **Magnesium:** Helps the body relax and pairs perfectly with Selank.
- **Avoid Alcohol:** Alcohol erases the benefits of this peptide.

Cautions:

- **Hair Shedding:** A small number of researchers report temporary hair shedding with long-term use (similar to Semax), though this stops when the cycle ends.
- **Drowsiness:** While rare, some people react to the "calm" by wanting to nap. First doses should always be taken at home.

Semaglutide

What it is / research focus: Semaglutide is a man-made copy of a natural hormone called GLP-1. Your body releases GLP-1 after you eat to tell your brain, "Okay, that's enough."

- **The Problem:** In many people, this natural signal is too quiet or goes away too fast, so they keep feeling hungry.
- **The Solution:** Semaglutide is a much louder, longer-lasting version of that signal.

How it works (The "Food Noise" Story): Imagine your brain is a radio that is constantly playing a song called *"I'm Hungry / What Should I Eat?"* This is called **"Food Noise."**

- **The Volume Knob:** For many people, this radio is stuck on maximum volume all day long.
- **The Action:** Semaglutide reaches up and turns the volume knob down to zero.
- **The Result:** The subject doesn't just feel "full"—they actually stop *thinking* about food. They can walk past a donut and not even care.

In simple terms: It turns off the "obsessive thoughts" about food in your brain and slows down your stomach so you stay full for hours.

The "Strange" Bonus: Researchers are noticing something weird: it doesn't just stop food cravings. Many people report they also stop wanting to **drink alcohol, smoke, or even shop online**. It seems to turn off the "craving switch" for many bad habits, not just eating!

Anecdotal:

- **The "Sulfur Burps":** Because food sits in the stomach for a long time, it can sometimes ferment, causing burps that smell like eggs. (Tip: Pepto-Bismol helps).
- **The "Freeze":** A very common but strange report is feeling freezing cold, even in a warm room. This is a sign the body is losing fat.

Beginner Protocol:

- **Dose:** 0.25 mg
- **Frequency:** Once per week (SubQ)
- **Cycle:** 4 weeks.
- **Important:** Do *not* rush this. Even if you feel nothing, you must let the body get used to it, or you will get very sick.

Intermediate Protocol:

- **Dose:** 0.5 mg
- **Frequency:** Once per week
- **Cycle:** 4 weeks.

Advanced Protocol:

- **Dose:** 1.0 mg – 2.4 mg
- **Frequency:** Once per week
- **Note:** Many researchers never need to go above 1.0 mg. If it is still working, stay at the lowest effective dose to save money and reduce side effects.

Supplements & diet:

- **The "Protein Rule":** You will not be hungry, so you might forget to eat. You *must* eat protein first, or you will lose muscle instead of fat.
- **Magnesium & Fiber:** This medicine slows down *everything*, including the bathroom. You need help to keep things moving.

Cautions:

- **Nausea:** This is the #1 enemy. It usually happens if you eat too much grease or sugar.
- **Muscle Loss:** If you don't eat enough protein, your body will eat your muscles for energy.

Semax

What it is / research focus: Semax is a legendary Russian "smart drug" originally developed by the Soviet Academy of Sciences. It was designed to keep astronauts and military personnel laser-focused during high-stress missions without making them jittery.

- **The "Ambulance" Fact:** In Russia, this is not just a study drug; it is an essential medicine carried in ambulances. It is used immediately after a stroke to stop brain damage and help neurons survive.

How it works (The Brain Fertilizer Story): Most stimulants (like caffeine or Adderall) work by shouting at your brain cells to work harder. Semax is different—it feeds them.

- **The Fertilizer (BDNF):** Semax rapidly increases a protein called **BDNF** (Brain-Derived Neurotrophic Factor). Think of BDNF as "Miracle-Gro" for your brain. It helps existing neurons survive and encourages new connections to grow.
- **The Shield:** It also breaks down enkephalins (natural painkillers), which helps the brain stay calm and clear under pressure.

In simple terms: It is "Brain Fertilizer." It helps you focus and learn faster by feeding your brain cells instead of just stimulating them.

Anecdotal: Researchers love Semax because it provides "clean" energy. There are no jitters, no crash, and no anxiety—just a feeling of effortless clarity.

- **The "Route" Debate:** While almost all clinical research uses **Nasal Drops** (because it travels directly from the nose to the brain), many biohackers prefer **SubQ injections** for a more potent, systemic effect.

Beginner Protocol:

- **Dose:** 0.1 mg – 0.3 mg (100 mcg – 300 mcg)
- **Frequency:** Daily
- **Method:** Nasal Spray (standard) or SubQ (advanced).
- **Cycle:** 2–4 weeks.

Intermediate Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg)
- **Frequency:** Daily
- **Note:** Most researchers take this in the morning, as it can be too stimulating for sleep.

Advanced Protocol:

- **Dose:** 1.0 mg – 1.5 mg (1,000 mcg – 1,500 mcg)
- **Frequency:** Daily
- **Cycle:** 4–6 weeks.

Supplements & diet:

- **Fish Oil (DHA/EPA):** This is non-negotiable. Semax increases the demand for building new brain cell connections, and Omega-3s are the raw material needed to build them.
- **Sleep:** Your brain physically repairs itself while you sleep; Semax makes this process more efficient.

Cautions:

- **Headaches:** The most common side effect. This is usually due to increased blood flow in the brain.
- **Hair Shedding:** A rare but persistent rumor in researcher communities suggests long-term, high-dose use might trigger temporary hair shedding (linked to BDNF levels), though this is not clinically proven.

Sermorelin

What it is / research focus: Sermorelin is a "secretagogue," which is a fancy word for a signal. It is a man-made version of the natural brain signal (GHRH) that tells your pituitary gland to release Growth Hormone.

How it works (The Night Shift Manager): Imagine your pituitary gland is a factory that makes Growth Hormone, but the workers are asleep.

- **The Problem:** As we get older, the "manager" (your natural GHRH) stops showing up to work, so the factory stops producing.
- **The Solution:** Sermorelin acts like a replacement manager. It walks into the factory right before you go to sleep and flips the "ON" switch.
- **The Result:** The factory workers wake up and make your *own* natural Growth Hormone while you sleep. This is different from injecting HGH, which is like ordering pizza instead of cooking—it's easier, but it makes the factory lazy.

In simple terms: It is a "Wake Up Call" for your pituitary gland. It tells your body to make its own

growth hormone again, naturally.

Anecdotal: This is the "Sleep & Skin" favorite. Almost every researcher reports the same timeline:

- **Weeks 1–2:** Deep, restorative sleep improves (vivid dreams are common).
- **Weeks 4–6:** Skin looks "glowy" or tighter; recovery from gym sessions gets faster.
- **Months 3–6:** Leaning out and fat loss becomes noticeable.
- *Note:* It is famous for causing a "histamine reaction"—a red, itchy bump at the injection site that looks like a mosquito bite. This usually goes away in 15–30 minutes.

Beginner Protocol:

- **Dose:** 0.1 mg – 0.2 mg (100 mcg – 200 mcg)
- **Frequency:** Daily (SubQ)
- **Timing: Strictly Pre-Bed.** Take it 30 minutes before sleep.

Intermediate Protocol:

- **Dose:** 0.2 mg – 0.3 mg (200 mcg – 300 mcg)
- **Frequency:** Daily
- **Timing:** Pre-Bed.

Advanced Protocol:

- **Dose:** 0.3 mg – 0.5 mg (300 mcg – 500 mcg)
- **Frequency:** Daily
- **Note:** Going higher than 500 mcg is usually a waste of money because the pituitary gland can only make so much at once (saturation point).

Supplements & diet:

- **The "Empty Stomach" Law:** This is critical. Insulin kills Sermorelin. You **must** wait at least 2 hours after your last meal to inject. If you eat right before bed, the shot will not work.
- **ZMA (Zinc & Magnesium):** Takes the sleep benefits to the next level.

Cautions:

- **The "Flush":** Many people feel a warm "rush" to their face for a minute after injecting. This is a good sign—it means it hit the bloodstream.
- **Injection Site:** As mentioned, red itchy bumps are very common and not an allergic reaction, just a local irritation.

SS-31 (Elamipretide)

What it is / research focus: SS-31 is a history-making peptide. In September 2025, it became the **first-ever mitochondria-targeted therapy to receive FDA approval** (under the brand name *Forzinity*) for treating Barth syndrome. It is a "cell-penetrating" peptide designed to repair the inner machinery of the cell's power plants (mitochondria). Unlike stimulants that just press the gas pedal, SS-31 fixes the engine.

How it works (The Loose Wire Story): Your mitochondria produce energy by passing electrons along a chain, like electricity flowing through a wire.

- **The Problem (Cardiolipin):** To keep this electricity flowing smoothly, the inner wall of the mitochondria needs a special fat called "Cardiolipin." Think of Cardiolipin as the **insulation** around the wire. As we age, this insulation gets frayed and slippery.
- **The "Leak":** When the insulation breaks, electrons leak out. This causes two problems: you lose energy (fatigue), and the leaking electricity sparks fires (oxidative stress/inflammation).
- **The Fix:** SS-31 acts like **Electrical Tape**. It binds directly to the frayed Cardiolipin, sealing the leak. This restores the flow of energy and stops the sparks.

In simple terms: It tightens the loose connections in your cells' batteries so they hold a charge better and don't overheat.

Anecdotal: Because it fixes the "source" of energy rather than acting like caffeine, researchers report a "clean" endurance boost. It is highly popular among endurance athletes (cyclists/runners) and the longevity community.

- **The "Sting":** The most consistent report is that SS-31 **burns** when injected. It is often mixed with BPC-157 or diluted to reduce the irritation.

Beginner Protocol:

- **Dose:** 2 mg – 4 mg
- **Frequency:** Daily (or 4 mg every other day)
- **Cycle:** 4–6 weeks
- *Note:* Consistent daily exposure is often preferred over large infrequent doses for mitochondrial repair.

Intermediate Protocol:

- **Dose:** 4 mg – 8 mg
- **Frequency:** Daily (or split doses)
- **Cycle:** 6–8 weeks

Advanced Protocol:

- **Dose:** 10 mg – 15 mg
- **Frequency:** Daily
- *Note:* Clinical trials for Barth syndrome used extremely high doses (up to 40mg daily), suggesting a very high safety ceiling, but cost usually limits researchers to lower ranges.

Supplements & diet:

- **Mitochondrial Stack:** Works best when paired with **CoQ10**, **PQQ**, and **NAD+** precursors (like Nicotinamide Riboside) to give the repaired mitochondria plenty of fuel.
- **Diet:** A diet rich in healthy fats (Omega-3s) supports the cell membranes.

Cautions:

- **Injection Pain:** This is famous for causing red, itchy, or stinging welts at the injection site. Rotating sites is mandatory.
- **FDA Note:** While approved for Barth syndrome, its use for anti-aging is still considered "off-label" research.

TB-500 (Thymosin β -4)

What it is / research focus: This is the synthetic version of a major healing protein that occurs naturally in humans, horses, and other mammals. While BPC-157 works on the gut and tendons, TB-500 is found in high concentrations in blood platelets and wound fluid.

How it works (The General Contractor Story): Think of your injury (a tear in the muscle or skin) as a construction site that has stalled because the workers can't move around.

- **The Problem:** The cells are stuck in traffic.
- **The Solution (Actin Sequestering):** TB-500 acts like a traffic controller. It changes the structure of the cell's skeleton (actin) so the cells become more flexible and mobile.
- **The Result:** Good cells can travel to the injury site much faster. It also builds new "roads" (blood vessels) to bring fresh nutrients to the damage.

In simple terms: It makes your cells "slippery" so they can slide quickly to where the injury is

and fix it. It also builds new blood vessels to feed the repair.

Anecdotal: This is often called the "Wolverine Peptide" in bodybuilder forums. Users report it works best for **muscle tears** and **flexibility**, whereas BPC-157 is favored for tendons/ligaments.

- **The "Loading" Phase:** Unlike most peptides, TB-500 is almost always "loaded" (high dose for 4–6 weeks) to saturate the tissue, followed by a much lower "maintenance" dose.

Beginner Protocol:

- **Load:** 2 mg – 4 mg per week (split into 2 shots, e.g., Monday/Thursday).
 - *Duration:* 4–6 weeks.
- **Maintenance:** 2 mg – 4 mg *once every 2 weeks*.

Intermediate Protocol:

- **Load:** 4 mg – 6 mg per week (split into 2 shots).
 - *Duration:* 4–6 weeks.
- **Maintenance:** 4 mg *once every 2 weeks*.

Advanced Protocol:

- **Load:** 6 mg – 8 mg per week (split into 2 or 3 shots).
 - *Duration:* 6–8 weeks.
- **Maintenance:** 4 mg – 6 mg *once every 2 weeks*.

Supplements & diet:

- **The Building Blocks:** Since you are building new tissue, you need raw materials. **Vitamin C** and **Collagen** are critical during the loading phase.
- **Protein:** Keep protein intake high to support muscle repair.

Cautions:

- **Water Retention:** This is the #1 reported side effect. Because it builds new blood vessels, the body holds extra water. This usually goes away when the loading phase ends.
- **Fatigue:** Many researchers report feeling surprisingly tired or lethargic on the day of injection.
- **The "Cancer" Question:** Because it helps blood vessels grow (angiogenesis), there is a theoretical risk that it could help existing tumors grow. Researchers with a history of cancer usually avoid this compound.

TB-500 Fragment (17–23)

What it is / research focus: This is a small, sliced-off piece of the larger Thymosin Beta-4 protein. Scientists found that this specific section (amino acids 17–23) is the "active site" responsible for fixing damaged tissue and preventing scar tissue.

How it works (The Car Key Story): Think of the full TB-500 (Thymosin Beta-4) molecule as a **Luxury Car**. It has a radio, air conditioning, and comfortable seats (immune system help, hair growth, etc.).

- **The Fragment:** This is just the **Key** to the car.
- **The Action:** The key starts the engine. It turns on the "healing mode" in the cells just like the full car does.
- **The Trade-off:** You get the engine power (tissue repair), but you don't get the radio or AC (other immune benefits). Because it is smaller, your body breaks it down much faster, so you have to "turn the key" (take a dose) more often.

In simple terms: The "Short-Cut" version of TB-500. It focuses only on fixing tears in skin and muscle but leaves the body very quickly.

Anecdotal: Researchers usually choose this for one of two reasons: it is often cheaper, or they only care about a specific muscle injury and don't need immune system support. It is almost always paired with **BPC-157**. Because it is so small, some reports claim it works better when injected closer to the injury (local), whereas the full TB-500 works anywhere (systemic).

Beginner Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg)
- **Frequency:** 3 times per week (SubQ)
- **Cycle:** 4–6 weeks

Intermediate Protocol:

- **Dose:** 1.0 mg – 1.5 mg (1,000 mcg – 1,500 mcg)
- **Frequency:** 3 times per week
- **Note:** Due to its short half-life, some researchers prefer smaller daily doses (e.g., 500 mcg daily) instead of larger doses 3x/week.

Advanced Protocol:

- **Dose:** 1.5 mg – 2.0 mg (1,500 mcg – 2,000 mcg)
- **Frequency:** 3 times per week
- **Cycle:** 6 weeks max

Supplements & diet:

- **Collagen & Vitamin C:** These are the "bricks" the body needs to rebuild the tissue this peptide is signaling to fix.
- **Hydration:** Essential for tissue elasticity.

Cautions:

- **Frequency Matters:** Because this fragment is "cleared" from the body so fast, missing a dose makes it much less effective than the full TB-500.
- **Confusion:** Many vials labeled "TB-500" are actually this fragment. It is important to know which one is being researched to get the dosing right.

Tesamorelin

What it is / research focus: Tesamorelin is a man-made version of a hormone your brain already uses. It is famous because it is the *only* peptide FDA-approved specifically to reduce "visceral fat" (the hard, dangerous fat around the stomach organs) in HIV patients.

How it works (The Sniper Story): Think of regular Growth Hormone as a "Shotgun"—it sprays energy everywhere, making muscles grow but also causing water retention and swelling.

- **The Specialist:** Tesamorelin is like a "Sniper."
- **The Target:** It travels to the brain and tells it to release Growth Hormone in a very specific pattern (pulsatile).
- **The Result:** This specific pattern seems to tell the body to burn stomach fat first, without the nasty side effects (like swollen wrists) that come from other growth hormone treatments.

In simple terms: The "Belly Fat Specialist." It tells your body to release natural growth hormone to burn deep stomach fat.

Anecdotal: This is considered the "Gold Standard" for fat loss by researchers, but it is expensive. The most common report is a significant tightening of the waistline after 2–3 months. Users often take it right before bed because it works best while sleeping.

Beginner Protocol:

- **Dose:** 1 mg (1,000 mcg)
- **Frequency:** Daily (SubQ)
- **Timing:** Take on an empty stomach (at least 2 hours after eating), usually right before bed.
- **Cycle:** 8–12 weeks

Intermediate Protocol (The Clinical Standard):

- **Dose:** 2 mg (2,000 mcg)
- **Frequency:** Daily
- **Note:** This is the dose used in most successful medical studies.

Advanced Protocol:

- **Dose:** 2 mg – 2.5 mg
- **Frequency:** Daily
- **Note:** Going higher than 2 mg rarely gives better results and costs a lot more money.

Supplements & diet:

- **The "Empty Stomach" Rule:** This peptide is fragile. If you inject it while there is food (insulin) in your blood, it will not work. You **must** be fasted.
- **Sleep:** It works by amplifying your body's natural night-time pulses, so good sleep is required.

Cautions:

- **The "Mosquito Bite":** Tesamorelin is known for leaving a red, itchy bump (like a bug bite) at the injection site for about 15 minutes. This is normal.
- **Blood Sugar:** While safer than HGH, it can still slightly raise blood sugar, so monitoring is smart.

Testagen (bioregulator)

What it is / research focus: Bioregulator associated with testicular tissue; posited to support peptide synthesis and antioxidant defenses locally.

In simple terms: A peptide mix aimed at supporting normal testicular function.

Anecdotal: Vitality notes.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Zinc-rich foods; sleep.

Cautions: Endocrine sensitivity considerations.

Thymagen (bioregulator)

What it is / research focus: Bioregulator linked to immune tissue; proposed to modulate T-cell function and inflammatory tone.

In simple terms: A peptide mix aimed at helping immune balance.

Anecdotal: Immune resilience discussions.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Vitamin D sufficiency; zinc.

Cautions: Monitor inflammatory responses.

Thymosin Alpha-1

What it is / research focus: Thymosin Alpha-1 is a natural peptide that your body already makes. It comes from a small organ in your chest called the thymus gland.

How it works (The Soldier School Story): Think of your immune system as an army of soldiers. Before these soldiers can go out and fight germs, they have to go to "Basic Training" to learn what to attack.

- **The School (The Thymus):** This is where the baby white blood cells go to learn.
- **The Teacher (Thymosin Alpha-1):** This peptide is the drill sergeant. It teaches the soldiers two things:
 1. **Attack the Bad Guys:** It helps them spot viruses and bacteria.
 2. **Don't Shoot the Good Guys:** It teaches them *not* to attack your own body (which

is what happens in allergies or autoimmune issues).

Why is this unique? Most immune boosters just scream "ATTACK!" which can make inflammation worse. Thymosin Alpha-1 is a "modulator"—it turns the volume up if it's too low, and turns it down if it's too loud.

In simple terms: It is the "Brain" of the immune system. It doesn't just make the immune system stronger; it makes it smarter so it knows exactly what to fight.

Anecdotal: This is known as the "Feel Good" peptide because side effects are almost non-existent for most researchers. It is often used during flu season or when a subject feels run down. It doesn't give a "buzz" or energy spike; the report is usually just "I didn't get sick."

Beginner Protocol:

- **Dose:** 0.5 mg – 1.0 mg (500 mcg – 1,000 mcg)
- **Frequency:** 2 times per week (SubQ)
- **Cycle:** 4–8 weeks

Intermediate Protocol:

- **Dose:** 1.0 mg – 1.5 mg
- **Frequency:** 2 times per week
- **Cycle:** 4–8 weeks

Advanced Protocol:

- **Dose:** 1.5 mg – 2.0 mg
- **Frequency:** 2 times per week
- **Cycle:** 4–8 weeks

Supplements & diet:

- **The Building Blocks:** Since the body is making new immune cells, it needs raw materials. Vitamin C, Vitamin D, and Zinc are the standard stack to help this peptide work.
- **Sleep:** The immune system repairs itself at night.

Cautions:

- **Very Safe:** Because this is natural to the body, it is considered one of the safest peptides to research.
- **Injection Site:** The only common complaint is a little red itchy spot where the needle went in.

Thymalin (bioregulator)

What it is / research focus: Thymic bioregulator referenced in historical literature; posited effects on immune tone and aging markers.

In simple terms: A peptide mix studied for immune balance and healthy aging.

Anecdotal: Immune tone.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Vitamin D; zinc.

Cautions: Limited modern trials.

Thyrotropin TRH (Protirelin)

What it is / research focus: Tripeptide releasing hormone used in thyroid axis testing; evokes TSH release for diagnostic purposes.

In simple terms: A quick test signal that tells the body to release thyroid-stimulating hormone.

Anecdotal: Acute testing only.

Research Protocol:

Beginner: 50–100 mcg SubQ single dose.

Intermediate: 100–200 mcg single dose.

Advanced: 200–400 mcg single dose.

Supplements & diet: None specific.

Cautions: Transient nausea/flushing.

Tirzepatide

What it is / research focus: This is currently considered the strongest weight-control medicine

available. While older medicines copy just one signal in the body, Tirzepatide copies *two* signals at the same time.

How it works (The "Two Button" Story): Imagine your body has two buttons that control how you handle food.

- **Button 1 (GLP-1):** Tells your brain "I am full" and tells your stomach to slow down so you don't get hungry again.
- **Button 2 (GIP):** Helps your body handle sugar and fat so they don't get stored as easily.
- **The Result:** Most medicines only push Button 1. Tirzepatide pushes **both buttons**. This makes the "stop eating" signal twice as strong.

In simple terms: It is a "Double Action" shot. It hits two targets instead of one to stop hunger and fix how the body uses sugar.

Anecdotal: Researchers often report this works much faster than anything else. The appetite suppression is very strong—sometimes too strong. The most common report is feeling "too full" or nauseous if the subject tries to eat a normal-sized meal.

Beginner Protocol:

- **Dose:** 2.5 mg – 5.0 mg
- **Frequency:** Once per week (SubQ)
- **Cycle:** Stay here for 8–12 weeks. Do not move up if it is still working.

Intermediate Protocol:

- **Dose:** 5.0 mg – 10 mg
- **Frequency:** Once per week
- **Note:** Only move up slowly (adding 2.5 mg at a time).

Advanced Protocol:

- **Dose:** 10 mg – 15 mg
- **Frequency:** Once per week
- **Note:** 15 mg is the maximum researched dose.

Supplements & diet:

- **Protein First:** Because appetite is so low, the subject might forget to eat. You *must* prioritize protein to stop muscle loss.
- **Hydration:** Drink water even if not thirsty to prevent headaches.

Cautions:

- **The "Sulfur Burps":** Because food sits in the stomach longer, it can cause burping or

indigestion.

- **Nausea:** If the dose is raised too fast, severe nausea is almost guaranteed. Slow and steady is the only way.

Vesilute (bioregulator)

What it is / research focus: Bioregulator often linked to bladder tissue in legacy references; posited effects on epithelial integrity.

In simple terms: A peptide mix aimed at bladder comfort.

Anecdotal: Urinary comfort notes.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Hydration.

Cautions: Limited modern trials.

Vesugen (bioregulator)

What it is / research focus: Vascular tissue-oriented bioregulator; proposed to support endothelial function and microcirculation.

In simple terms: A peptide mix aimed at helping blood vessels stay healthy. **Anecdotal:** Circulation comfort notes.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Cardio-friendly diet.

Cautions: As above.

Vilon (bioregulator)

What it is / research focus: Thymic/immune-related bioregulator in older literature; posited to support peptide synthesis and immune balance.

In simple terms: A peptide mix studied for general immune support.

Anecdotal: General vitality notes.

Research Protocol:

Beginner: 10–20 mg SubQ weekly · **Cycle:** 4–6 weeks.

Intermediate: 20–30 mg weekly.

Advanced: 30–40 mg weekly.

Supplements & diet: Vitamin D, zinc.

Cautions: Limited modern trials.

VIP (Vasoactive Intestinal Peptide)

What it is / research focus: Neuro-gastrointestinal peptide studied for smooth muscle relaxation and immune signaling; also examined in respiratory pathways.

In simple terms: A body peptide that relaxes muscles in the gut and lungs.

Anecdotal: Respiratory comfort notes.

Research Protocol:

Beginner: 50–100 mcg SubQ 2–3×/week · **Cycle:** 4–6 weeks.

Intermediate: 100–200 mcg 2–3×/week.

Advanced: 200–400 mcg 2–3×/week.

Supplements & diet: Hydration; sleep.

Cautions: Flushing, hypotension-like sensations possible.